

Advanced Macroeconomics II

Lecture Notes

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Chapter 1

Introduction to Dynamic Analysis

1.1 Dynamic Optimization

1.1.1 Convex Function

We begin by introducing the most fundamental concept in convex analysis: the convex set and the convex function. Here we define an operator $\phi(x, y) = \alpha x + (1 - \alpha)y$, where $\alpha \in [0, 1]$.

Definition 1.1.1: Convex Set

$A \subset \mathbb{X}$ is a convex set, if $\forall x, y \in A, \forall \alpha \in [0, 1], \phi(x, y) \in A$.

Definition 1.1.2: Convex Function

$f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a convex function, if $\text{dom}(f)$ is a convex set and $\forall x, y \in \text{dom}(f), \forall \alpha \in [0, 1]$, there have $f(\phi(x, y)) \leq \phi(f(x), f(y))$. (strictly convex if $f(\phi(x, y)) \neq \phi(f(x), f(y))$).

The convex function admits the following two equivalent conditions.

Proposition 1.1.3

If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable and $\text{dom}(f)$ is a convex set, then f is a convex function iff $\forall x, y \in \text{dom}(f), f(y) \geq f(x) + \nabla f(x)^T(y - x)$.

Proof. Necessity: Suppose f is convex, then $\forall x, y \in \text{dom}(f), \forall \alpha \in (0, 1]$, we have $f(x + \alpha(y - x)) \leq f(x) + \alpha(f(y) - f(x))$, i.e.

$$\frac{f(x + \alpha(y - x)) - f(x)}{\alpha} \leq f(y) - f(x).$$

Then taking $\alpha \rightarrow 0^+$ and yields

$$\lim_{\alpha \rightarrow 0^+} \frac{f(x + \alpha(y - x)) - f(x)}{\alpha} = \nabla f(x)^T(y - x) \leq f(y) - f(x).$$

Sufficiency: Denote $z = \phi(x, y)$. Suppose the condition holds, $\forall x, y \in \text{dom}(f)$, $\forall \alpha \in [0, 1]$, we have

$$\begin{aligned} f(x) &\geq f(z) + \nabla f(z)^T(x - z), \\ f(y) &\geq f(z) + \nabla f(z)^T(y - z). \end{aligned}$$

We obtain

$$\phi(f(x), f(y)) \geq f(z) + \nabla f(z)^T[\alpha(x - z) + (1 - \alpha)(y - z)] = f(z) = f(\phi(x, y)).$$

Thus, f is convex. □

Proposition 1.1.4

If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is twice differentiable and $\text{dom}(f)$ is a convex set, then f is a convex function iff $\forall x \in \text{dom}(f)$, $\nabla^2 f(x) \succeq 0$ i.e., the Hessian is positive semi-definite.

Proof. Necessity: Suppose f is convex, take the second-order Taylor expansion to f at x , we obtain

$$f(x + \delta x) = f(x) + \nabla f(x)^T \delta x + \frac{1}{2} \delta x^T \nabla^2 f(x) \delta x + R(x; \delta x) \|\delta x\|^2,$$

where $R(x; \delta x) \rightarrow 0$ as $\delta x \rightarrow 0$.

Since $f(x + \delta x) \geq f(x) + \nabla f(x)^T \delta x$, $\forall \delta x$ we have

$$\delta x^T \nabla^2 f(x) \delta x + R(x; \delta x) \|\delta x\|^2 \geq 0.$$

Let $\delta x = \varepsilon d$, taking $\varepsilon \rightarrow 0$ and yields $d^T \nabla^2 f(x) d \geq 0$, $\forall d$. Thus $\nabla^2 f(x) \succeq 0$.

Sufficiency: Denote $z = \phi(x, y)$. Suppose the condition holds, $\forall x, y \in \text{dom}(f)$, $\forall \alpha \in [0, 1]$, we have

$$\begin{aligned} f(y) &= f(x + (y - x)) \\ &= f(x) + \nabla f(x)^T(y - x) + \frac{1}{2}(y - x)^T \nabla^2 f(z)(y - x) \\ &\geq f(x) + \nabla f(x)^T(y - x). \end{aligned}$$

By Proposition 1.1.3, f is convex. □

For a unconstrained optimization problem:

$$\inf_{x \in \mathbb{R}^n} f(x).$$

By Proposition 1.1.3, if $\nabla f(x) = 0$, then $\forall y \in \text{dom}(f)$, $f(y) \geq f(x)$. It shows that such x is the minimum point.

1.1.2 Lagrangian Method

Now we turn to discuss the constrained optimization problem:

$$\inf_{x \in \mathbb{R}^n} f(x),$$

s.t.

$$\begin{aligned} g_i(x) &\leq 0, \quad i = 1, \dots, m, \\ h_j(x) &= 0, \quad j = 1, \dots, p \end{aligned}$$

Denote x^* as the minimum point and p^* as the solution, i.e., $p^* = \inf_{x \in \Gamma(\cdot)} f(x^*)$.

Then We define the Lagrangian of the primal problem:

$$\mathcal{L}(x, \lambda, \varphi) = f(x) + \lambda g + \varphi h,$$

where (λ, φ) are the Lagrange multipliers associated with constraints, and $\lambda > 0$.

The dual function is defined as $\psi(\lambda, \varphi) = \inf_{x \in \mathbb{X}} \mathcal{L}(x, \lambda, \varphi)$, and we denote the minimum point $x = \pi(\lambda, \varphi)$. By taking the infimum simultaneously on both sides of the equality, we obtain $\psi(\lambda, \varphi) \leq p^*$, in other words, p^* is the upper bound of $\psi(\lambda, \varphi)$. So the dual problem can be represented as

$$d^* = \sup_{(\lambda, \varphi) \in \mathbb{R}^m \times \mathbb{R}^p} \psi(\lambda, \varphi).$$

We call $d^* - p^* \leq 0$ the duality gap, and if $d^* - p^* = 0$, then we say the strong duality of dual problem holds. To ensure the solution implied from a dual problem is equivalent to that of primal problem, one shall require the strong duality.

If f, g, h are all differentiable, then a first-order necessary condition for optimality, as known as Karash-Kuhn-Tacker condition, are given by

$$\begin{aligned} \nabla f(x) + \nabla g(x) \cdot \lambda + \nabla h(x) \cdot \varphi &= 0, \\ \lambda \cdot g(x) &= 0, \quad g(x) \leq 0, \quad h(x) = 0. \end{aligned}$$

1.1.3 Optimal Control

See [Acemoglu \(2007\)](#) chapter 7.

1.2 Application: Basic Ramsey-Cass-Koopmans Model

1.2.1 Environment

We introduce a basic deterministic Ramsey-Cass-Koopmans model in discrete time, which has no government sector, but population growth and technological growth. In addition, we only considered a sequential trading asset market and the notion of sequential competitive equilibrium.

For households, there is a continuum of households of measure unity. Each household consists of identical members whose size grows at gross rate g_n . Household derives utility from consumption according to the following function:

$$\sum_{t=0}^{\infty} (\beta g_n)^t u(c_t),$$

where $\beta \in (0, 1)$ is the subjective discount factor and c_t represent a household member's consumption. To ensure the solution properties of the decision problem, we assume $\frac{\partial u}{\partial c} > 0$, $\frac{\partial^2 u}{\partial c^2} < 0$ and $\lim_{c \rightarrow 0} u(c, \cdot) = \infty$.

Households receive income from firms by renting capital and supplying labor. Then a representative household's budget constraint is given by

$$c_t + i_t = R_t k_t + w_t n,$$

where i_t , k_t , R_t , and w_t represent investment, capital, the rental rate, and the wage rate, respectively.

Naturally, the dynamic transfer of assets should satisfy:

$$g_n k_{t+1} = (1 - \delta) k_t + i_t,$$

where $\delta \in (0, 1)$ is the depreciation rate of capital.

By the FOCs, we obtain the Euler equation of household's UMP:

$$\frac{u_c(c_{t+1})}{u_c(c_t)} = \frac{1}{\beta(R_{t+1} + 1 - \delta)}.$$

Moreover, we impose the transversality condition as follows:

$$\lim_{t \rightarrow \infty} \beta^t u_c(c_t, n_t) k_{t+1} = 0.$$

Combined the FOCs with two constraints, one has derived the optimal decision concerning $\{c_t, k_{t+1}, i_t\}_{t=0}^{\infty}$, under the given sequences of market price R_t .

For firms, there is a continuum of identical firms of measure unity. Each firm combines capital and labor to produce goods using a CRS production function. Technology A_t growth at the gross rate g_a so that $A_{t+1} = g_a A_t$. The aggregate production is given by

$$Y_t = F(K_t, A_t L_t),$$

where Y_t , K_t , and L_t represent aggregate output, capital, and labor, respectively. Assume that F is homogeneous of degree one and satisfies $F(0, \cdot) = F(\cdot, 0) = F(0, 0) = 0$, $\frac{\partial F}{\partial K} > 0$, $\frac{\partial F}{\partial L} > 0$, $\frac{\partial^2 F}{\partial K^2} < 0$, $\frac{\partial^2 F}{\partial L^2} < 0$, and Inada conditions.

Since the firm rents capital and hires labor each period, it solves the following static problem¹:

$$\Pi_t = \max_{K_t, L_t} F(K_t, A_t L_t) - R_t K_t - w_t L_t.$$

¹The price of the output is normalized as 1.

Its optimal solution implies,

$$\begin{aligned}\frac{\partial F(K_t, A_t L_t)}{\partial K_t} &= R_t, \\ \frac{\partial F(K_t, A_t L_t)}{\partial L_t} &= w_t.\end{aligned}$$

In this model, a sequential competitive equilibrium consists sequences of per capita quantities $\{c_t, k_{t+1}, i_t\}_{t=0}^{\infty}$, aggregate quantities $\{C_t, I_t, L_t, K_{t+1}\}_{t=0}^{\infty}$, and price $\{w_t, R_t\}_{t=0}^{\infty}$ such that:

1. Given prices, $\{c_t, k_{t+1}, i_t\}_{t=0}^{\infty}$ solve the UMP.
2. Given prices, $\{K_t, L_t\}$ solve the PMP.
3. All markets clear, in the sense that $C_t + I_t = F(K_t, A_t L_t)$ or in the form of per capita $c_t + i_t = F(k_t, A_t \bar{l})$.

To discuss the efficiency of equilibrium, we consider a social planner's problem in which the social planner chooses consumption, and investment to maximize a representative household's utility subject to the resource constraint.

Specifically, social planner chooses c_t , i_t , and k_{t+1} so that solve the following problem:

$$\max_{c_t, i_t} \sum_{t=0}^{\infty} \beta^t N_t u(c_t, l_t)$$

s.t.

$$\begin{aligned}N_t(c_t + i_t) &= F(N_t k_t, A_t N_t \bar{l}), \\ N_t k_{t+1} &= (1 - \delta) N_{t-1} k_t + N_t i_t.\end{aligned}$$

The Euler equation is given by

$$\frac{u_c(c_{t+1})}{u_c(c_t)} = \frac{1}{\beta(F_k(k_t, A_{t+1} \bar{l}) + 1 - \delta)}.$$

It shows that the allocations under the social planner is the same as that of the decentralized equilibrium. In other words, the allocations and prices in this model reached the Pareto optimality.

1.2.2 Solution

In this section, we use a CIES utility function $u(c_t) = \frac{c_t^{1-\sigma}}{1-\sigma}$ where $\sigma > 0$ and $\sigma \neq 1$, and a Cobb-Douglas production function $F(K_t, A_t L_t) = K_t^\alpha (A_t L_t)^{1-\alpha}$, where $\alpha \in (0, 1)$.

The complete economic dynamic system can be expressed as

$$\begin{aligned}u_c(c_t) &= \beta u_c(c_{t+1}) [F_k(k_t, A_{t+1} \bar{l}) + 1 - \delta], \\ N_t(c_t + i_t) &= F(N_t k_t, A_t N_t \bar{l}), \\ N_t k_{t+1} &= (1 - \delta) N_{t-1} k_t + N_t i_t.\end{aligned}$$

By de-trending the population growth and technical progress $N_t A_t$, we obtain a stable dynamic system:

$$\begin{aligned} u_c(\tilde{c}_t) &= \beta g_a^{-\sigma} u_c(\tilde{c}_{t+1}) [F_k(\tilde{k}_{t+1}, \bar{l}) + 1 - \delta], \\ \tilde{c}_t + \tilde{i}_t &= F(\tilde{k}_t, \bar{l}), \\ g_n g_a \tilde{k}_{t+1} &= (1 - \delta) \tilde{k}_t + \tilde{i}_t, \end{aligned}$$

where $\tilde{c}_t = \frac{c_t}{N_t A_t}$, $\tilde{k}_t = \frac{k_t}{N_t A_t}$, and $\tilde{i}_t = \frac{i_t}{N_t A_t}$.

In steady state, the fixed point satisfies $\tilde{c}_t = \tilde{c}_{t+1} = \bar{c}$, $\tilde{k}_t = \tilde{k}_{t+1} = \bar{k}$, and the system can be represented as

$$\begin{aligned} 1 &= \beta g_a^{-\sigma} [F_k(\bar{k}, \bar{l}) + 1 - \delta], \\ g_n g_a \bar{k} &= (1 - \delta) \bar{k} + F(\bar{k}, \bar{l}) - \bar{c}, \end{aligned}$$

where $F(\bar{k}, \bar{l}) = \bar{k}^\alpha \bar{l}^{1-\alpha}$.

We derive the steady state of capital and consumption as

$$\begin{aligned} \bar{k} &= \left[\frac{g_a^\sigma \beta^{-1} - (1 - \delta)}{\alpha} \right]^{\frac{1}{\alpha-1}} \cdot \bar{l}, \\ \bar{c} &= (1 - \delta - g_n g_a) \bar{k} + F(\bar{k}, \bar{l}). \end{aligned}$$

Then we consider the log-linear approximation of this system:

$$\begin{aligned} \hat{k}_{t+1} &= \frac{(1 - \delta) \bar{k} + \alpha F(\bar{k}, \bar{l})}{g_n g_a \bar{k}} \hat{k}_t - \frac{\bar{c}}{g_n g_a \bar{k}} \hat{c}_t, \\ \hat{c}_{t+1} &= \hat{c}_t + \frac{\beta \alpha (\alpha - 1)}{\sigma} \frac{F(\bar{k}, \bar{l})}{\bar{k}} \hat{k}_{t+1}, \end{aligned}$$

where $\hat{c}_t = \ln \tilde{c}_t - \ln \bar{c} \approx \frac{\tilde{c}_t - \bar{c}}{\bar{c}}$, and $\hat{k}_t = \ln \tilde{k}_t - \ln \bar{k} \approx \frac{\tilde{k}_t - \bar{k}}{\bar{k}}$.

When the growth rate $g_n = g_a = 1$, the log-linearized dynamic system can be simplified as

$$\begin{aligned} \Delta \hat{k}_t &= \hat{k}_{t+1} - \hat{k}_t = (\beta^{-1} - 1) \hat{k}_t - \frac{\bar{c}}{\bar{k}} \hat{c}_t, \\ \Delta \hat{c}_t &= \hat{c}_{t+1} - \hat{c}_t = \frac{\beta \alpha (\alpha - 1)}{\sigma} \frac{F(\bar{k}, \bar{l})}{\bar{k}} \hat{k}_{t+1}. \end{aligned}$$

Qualitatively, the system shows that $\frac{\partial \Delta \hat{k}_t}{\partial \hat{k}_t} > 0$, $\frac{\partial \Delta \hat{k}_t}{\partial \hat{c}_t} < 0$, and $\frac{\partial \Delta \hat{c}_t}{\partial \hat{k}_t} < 0$. Consider the isocline for two variables, we obtain

$$\begin{aligned} \Delta \hat{c}_t = 0 &\Rightarrow \hat{k}_t = 0, \\ \Delta \hat{k}_t = 0 &\Rightarrow \hat{k}_t = \frac{1}{\beta^{-1} - 1} \frac{\bar{c}}{\bar{k}} \hat{c}_t. \end{aligned}$$

Then the phase diagram can be showed as

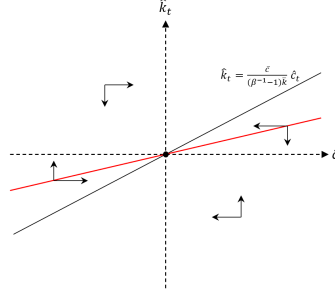


Figure 1.1: Phase diagram of basic Ramsey model

1.3 Application: Dynamic Input-Output Network Model

Comparing to the static input-output models ([Acemoglu et al., 2012](#)), the dynamic version shows how temporary shocks affects persistently due to adjustment costs in input-output linkages.

In this section, We extend the social planner's problem by [Liu and Tsyvinski \(2024\)](#) to a decentralized economy.

1.3.1 Environment

In this continuous-time model, there is a representative consumer with exogenous labor supply $\bar{l}$ and N production sectors that produce from labor and intermediate inputs. The consumer has utility

$$V \equiv \int_0^\infty e^{-\rho t} \ln c(t) dt,$$

where $c(t)$ is a Cobb-Douglas aggregator over sectoral goods $j = 1, \dots, N$:

$$c(t) = \prod_{j=1}^N c_j(t)^{\beta_j}, \quad \sum_{j=1}^N \beta_j = 1,$$

where $\beta_i > 0$. By the property of Cobb-Douglas utility, the demand function becomes

$$p_j(t)c_j(t) = \beta_j Z(t), \tag{1.3.1}$$

where $p_j(t)$ is the price of good j , $Z(t)$ is the total expenditure.

At each time t , the output of production i satisfies

$$q_i(t) = z_i(t)l_i(t)^{\alpha_i} \prod_{j=1}^N m_{ij}(t)^{\sigma_{ij}}, \quad \alpha_i + \sum_{j=1}^N \sigma_{ij} = 1,$$

where $0 < \alpha_i, \sigma_{ij} < 1$, $z_i(t)$ is TFP, $l_i(t)$ is the amount of labor used, and $m_{ij}(t)$ is the amount of the intermediate good of the sector j used in the production of the good i .

To use input m_{ij} at time t , firm i needs to buy

$$s_{ij} = m_{ij} \times \exp\left(\delta \frac{\dot{m}_{ij}}{m_{ij}}\right)$$

units of input j . The term $\exp\left(\delta \frac{\dot{m}_{ij}}{m_{ij}}\right)$ represents the adjustment cost that producer i has to incur when it raises the quantity of input j . The logarithmic form becomes

$$\dot{m}_{ij}(t) = \delta^{-1} m_{ij}(t) [\ln s_{ij}(t) - \ln m_{ij}(t)].$$

Firm i 's intertemporal problem can be represented as

$$\max_{l_i(\cdot), \{s_{ij}(\cdot)\}} \int_0^\infty e^{-\rho t} \left[p_i(t) q_i(t) - w(t) l_i(t) - \sum_{j=1}^N p_j(t) s_{ij}(t) \right] dt.$$

The goods and labor market clearing conditions are

$$q_j(t) = c_j(t) + \sum_{i=1}^N s_{ij}(t) \quad \forall j, \quad (1.3.2)$$

$$\sum_{j=1}^N l_j(t) = \bar{l}. \quad (1.3.3)$$

For simplicity, we assume goods delivered to the consumer are not subject to adjustment costs, and neither is the use of labor across production sectors.

From now on, wherever it does not cause confusion, we suppress dependence on time t in the notation.

Consider the current-value Hamiltonian for firm i :

$$\begin{aligned} \mathcal{H}_i &= p_i q_i - w l_i - \sum_{j=1}^N p_j s_{ij} + \sum_{j=1}^N \mu_{ij} \dot{m}_{ij} \\ &= p_i q_i - w l_i - \sum_{j=1}^N p_j s_{ij} + \sum_{j=1}^N \mu_{ij} \delta^{-1} m_{ij} (\ln s_{ij} - \ln m_{ij}), \end{aligned}$$

where μ_{ij} denotes the shadow value of m_{ij} . The FOCs of control variables become

$$\frac{\partial \mathcal{H}_i}{\partial l_i} : w = p_i \alpha_i \frac{q_i}{l_i}, \quad (1.3.4)$$

$$\frac{\partial \mathcal{H}_i}{\partial s_{ij}} : \mu_{ij} = \delta p_j \frac{s_{ij}}{m_{ij}}. \quad (1.3.5)$$

For the state variable, we obtain the Euler equation of μ_{ij} by FOC:

$$\begin{aligned} \dot{\mu}_{ij} &= \rho \mu_{ij} - \frac{\partial \mathcal{H}_i}{\partial m_{ij}} \\ &= \rho \mu_{ij} - \left[p_i \sigma_{ij} \frac{q_i}{m_{ij}} + \mu_{ij} \delta^{-1} (\ln s_{ij} - \ln m_{ij} - 1) \right]. \end{aligned}$$

This is a two-dimensional ODE system about both μ_{ij} and m_{ij} , which is difficult to be solved directly. Thus, we define $\zeta_{ij} \equiv \mu_{ij} m_{ij}$ as the value of input stock. By 1.3.5, we have $\dot{\zeta}_{ij} = \delta p_j s_{ij}$.

Consider the derivative of ζ_{ij} with respect to t :

$$\begin{aligned}\dot{\zeta}_{ij} &= \dot{\mu}_{ij}m_{ij} + \mu_{ij}\dot{m}_{ij} \\ &= \left[\rho\mu_{ij} - p_i\sigma_{ij}\frac{q_i}{m_{ij}} - \mu_{ij}\delta^{-1}(\ln s_{ij} - \ln m_{ij} - 1) \right] m_{ij} + \mu_{ij}\delta^{-1}m_{ij}(\ln s_{ij} - \ln m_{ij}) \\ &= \rho\mu_{ij}m_{ij} - p_i\sigma_{ij}q_i + \mu_{ij}m_{ij}\delta^{-1} \\ &= (\rho + \delta^{-1})\zeta_{ij} - p_i\sigma_{ij}q_i,\end{aligned}$$

which is a linear one-dimensional ODE system.

Recall the transversality condition:

$$\lim_{t \rightarrow \infty} e^{-\rho t} \mu_{ij}(t) m_{ij}(t) = \lim_{t \rightarrow \infty} e^{-\rho t} \zeta_{ij}(t) = 0.$$

Let $t = 0$, it becomes

$$\lim_{t \rightarrow \infty} e^{-\rho t} \zeta_{ij}(0) = 0,$$

which implies $\zeta_{ij}(0) = 0$ as the initial condition.

Then solving the ODE system:

$$\zeta_{ij}(t) = e^{(\rho + \sigma^{-1})t} \zeta_{ij}(0) - \int_0^t e^{(\rho + \sigma^{-1})(t-s)} p_i(s) \sigma_{ij} q_i(s) ds.$$

In equilibrium, economy converges to steady state, i.e, $\dot{\zeta}_{ij} = 0$. Hence,

$$(\rho + \sigma^{-1})\zeta_{ij} = (\rho + \sigma^{-1})\delta p_j s_{ij} = p_i \sigma_{ij} q_i,$$

which can be simplified as

$$p_j s_{ij} = \frac{p_i \sigma_{ij} q_i}{1 + \rho \delta}. \quad (1.3.6)$$

In this model, firm i 's demand for input j depends on its own output value $p_i q_i$, the input elasticity σ_{ij} , the adjustment cost δ and the discount rate ρ .

1.3.2 Proportionality Relations

Lemma 1.3.1

For any initial condition $\{m_{ij}(0)\}$, the solution to the problem features a constant fraction of each good j being sent to the consumer and to each input user i along the entire transitional path: $\forall i, j$, $\frac{c_j(t)}{q_j(t)}$ and $\frac{s_j(t)}{q_t(t)}$ are time invariant. The solution also features a constant labor allocation along the entire transition path as $\forall j$, $\frac{l_i(t)}{l}$ is also time invariant.

Proof. Define the intermediate input sales proportion by $v_{ij} \equiv \frac{s_{ij}}{q_j}$, and the consumption proportion by $\omega_j = \frac{c_j}{q_j}$. By 1.3.3, we have

$$\omega_j + \sum_{i=1}^N v_{ij} = 1 \quad \forall j, \quad (1.3.7)$$

and by 1.3.1, we have

$$p_j q_j = \frac{\beta_j Z}{\omega_j}. \quad (1.3.8)$$

From 1.3.6, dividing both sides by $p_j q_j$ yields

$$\frac{s_{ij}}{q_j} = \frac{p_i \sigma_{ij} q_i}{p_j q_j (1 + \rho \sigma)},$$

then by 1.3.7 we obtain

$$v_{ij} \equiv \frac{s_{ij}}{q_j} = \frac{\sigma_{ij} \beta_i \omega_j}{(1 + \rho \sigma) \beta_j \omega_i}. \quad (1.3.9)$$

Substituting 1.3.9 into 1.3.7, we obtain

$$\omega_j + \sum_{i=1}^N \frac{\sigma_{ij} \beta_i \omega_j}{(1 + \rho \delta) \beta_j \omega_i} = 1,$$

then solving for ω_j :

$$\omega_j = \left(1 + \frac{1}{\beta_j} \sum_{i=1}^N \frac{\sigma_{ij} \beta_i}{(1 + \rho \sigma) \omega_i} \right)^{-1} \quad (1.3.10)$$

In equilibrium, the existence of a solution requires ω_j to be constant, ensuring proportionality invariance along the transition path.

Now we consider the labor allocation proportionality. Combining 1.3.3 and 1.3.4:

$$\frac{l_i}{\bar{l}} = \frac{p_i \alpha_i q_i}{\sum_k p_k \alpha_k q_k},$$

then by 1.3.8 we have

$$\frac{l_i}{\bar{l}} = \frac{\alpha_i \beta_i \omega_i^{-1}}{\sum_k \alpha_k \beta_k \omega_k^{-1}}.$$

When w_i is constant, the labor allocation proportionality $\frac{l_i}{\bar{l}}$ is proportional to $\alpha_i \beta_i$, which is constant. $\square$

Define

$$x_j(t) \equiv \ln \sum_{i=1}^N s_{ij}(t) - \ln \sum_{i=1}^N m_{ij}(t),$$

which is the log-ratio between the quantity of good j supplied to and used by other producers. In a steady state, $\frac{\sum_j s_{ij}}{\sum_i m_{ij}} = 1$, then $\forall j, x_j = 0$. Away from a steady state, the ratio captures the proportional adjustment costs incurred for expanding input j in production. One can prove that both $m_{ij}(t)$ and $s_{ij}(t)$ grow at rates independent of i during the transition, so $\delta^{-1} x_j(t)$ is the rate at which all sectors expand their use of input j .

1.3.3 Solution

In what follows, we use boldface to denote vectors (lower case) and matrices (upper case). Let $\Sigma \equiv [\sigma_{ij}]$ denote the matrix of input-output elasticities, and let β denote the $N \times 1$ vector of consumption elasticities. Let α be the vectore of sectoral value-added shares. Let $\gamma' \equiv \beta'(\mathbf{I} - \Sigma)^{-1}$ be the vector of Domar weights.

Lemma 1.3.2: Laws of Motion for Sectoral Output and GDP

Consider the economy recovering at time 0 from a TFP shock vector $\tilde{\mathbf{z}}$, then

1. The law of motion for sectoral output $\mathbf{q}$ is

$$\frac{d \ln \mathbf{q}}{dt} = \delta^{-1} \mathbf{\Sigma} \mathbf{x}(t), \quad (1.3.11)$$

with initial condition $\ln \mathbf{q}(0) = \ln \mathbf{q}^{ss} - \beta'(\mathbf{I} - \mathbf{\Sigma})^{-1} \tilde{\mathbf{z}}$.

2. The law of motion for GDP is

$$\frac{d \ln c(t)}{dt} = \delta^{-1} \beta' \mathbf{\Sigma} \mathbf{x}(t), \quad (1.3.12)$$

with initial condition $\ln c(0) = \ln c^{ss} - \beta'(\mathbf{I} - \mathbf{\Sigma})^{-1} \tilde{\mathbf{z}}$.

3. The path of the log-ratio between inputs supplied and used satisfies

$$\frac{d \mathbf{x}(t)}{dt} = -\delta^{-1} (\mathbf{I} - \mathbf{\Sigma}) \mathbf{x}(t), \quad (1.3.13)$$

with initial condition $\mathbf{x}(0) = \tilde{\mathbf{z}}$.

Proof. Since all sectors uses input j with same growth rate, we have

$$\frac{d}{dt} \ln \left(\sum_{i=1}^N m_{ij} \right) = \delta^{-1} x_j(t).$$

By the production function, the output growth rate for sector j becomes

$$\frac{\dot{q}_j}{q_j} = \frac{\dot{z}_j}{z_j} + \alpha_j \frac{\dot{l}_j}{l_j} + \sum_{k=1}^N \sigma_{jk} \frac{\dot{m}_{jk}}{m_{jk}}.$$

During recovery, $\dot{z}_j = \dot{l}_j = 0$, thus,

$$\frac{\dot{q}_j}{q_j} = \sum_{k=1}^N \sigma_{jk} \frac{\dot{m}_{jk}}{m_{jk}} = \delta^{-1} \sum_{k=1}^N \sigma_{jk} x_k(t).$$

Transforming this equation into matrix form, 1.3.11 is proved.

Since constant fraction of output allocated to intermediate inputs, we have

$$\frac{d}{dt} \ln \left(\sum_{i=1}^N s_{ij} \right) = \frac{d \ln q_j}{dt} = \delta^{-1} \sum_{k=1}^N \sigma_{jk} x_k(t).$$

By definition of $x_j(t)$,

$$\begin{aligned}\frac{dx_j(t)}{dt} &= \frac{d}{dt} \ln \left(\sum_{i=1}^N s_{ij} \right) - \frac{d}{dt} \ln \left(\sum_{i=1}^N m_{ij} \right) \\ &= -\delta^{-1} \left[x_j(t) - \sum_{k=1}^N \sigma_{jk} x_k(t) \right].\end{aligned}$$

Transforming this equation into matrix form, [1.3.13](#) is proved.

Since constant fraction β_j of each good j sent to final consumption, taking logarithms and differentiating the consumption function:

$$\frac{d \ln c(t)}{dt} = \sum_{j=1}^N \beta_j \frac{d \ln c_j(t)}{dt} = \sum_{j=1}^N \beta_j \frac{d \ln q_j(t)}{dt}.$$

By [1.3.11](#), [1.3.12](#) is proved. □

Finally, we obtain the solution to the dynamic system.

By [1.3.13](#), we have

$$\mathbf{x}(t) = \exp[-\delta^{-1}(\mathbf{I} - \mathbf{\Sigma})t] \tilde{\mathbf{z}}. \quad (1.3.14)$$

By [1.3.11](#) and [1.3.14](#), we have

$$\begin{aligned}\ln \mathbf{q}(t) &= \ln \mathbf{q}(0) + \delta^{-1} \mathbf{\Sigma} \int_0^t \mathbf{x}(s) ds \\ &= \ln \mathbf{q}^{ss} - \beta' \mathbf{\Sigma} (\mathbf{I} - \mathbf{\Sigma})^{-1} \exp[-\delta^{-1}(\mathbf{I} - \mathbf{\Sigma})t] \tilde{\mathbf{z}}.\end{aligned}$$

Similarly for GDP:

$$\ln \mathbf{c}(t) = \ln \mathbf{c}^{ss} - \beta' \mathbf{\Sigma} (\mathbf{I} - \mathbf{\Sigma})^{-1} \exp[-\delta^{-1}(\mathbf{I} - \mathbf{\Sigma})t] \tilde{\mathbf{z}}.$$

Chapter 2

Deterministic Difference Systems

2.1 Scalar Linear Difference Equations

2.1.1 First-Order Linear Difference Equations

Consider the following scalar first-order linear difference equation:

$$x_{t+1} = bx_t + cz_t, \quad t \geq 0,$$

where x_t , b , c , and z_t are all real numbers. Assume that $\{z_t\}_{t=0}^{\infty}$ is an exogenously given bounded sequence. If $cz_t = 0$, $\forall t$, then this equation is homogeneous. And if z_t is constant for each t , then it is autonomous.

In the autonomous case, we may suppose $z_t = 1$, $\forall t$ without loss of generality. We then obtain:

$$x_{t+1} = bx_t + c.$$

Under the initial value, we iterate the above dynamic equation forward from period 0:

$$x_t = b^t \left(x_0 - \frac{c}{1-b} \right) + \frac{c}{1-b}.$$

- If $|\beta| < 1$, then the solution will always converge to the stationary point $\bar{x} = \frac{c}{1-b}$. In this case, we call $\bar{x}$ a globally asymptotically stable steady state. When x_0 is not given, then the solution is indeterminate.
- If $|\beta| > 1$, then the solution explodes or is unstable, $\forall x_0 \neq \bar{x}$, unless $x_0 = \bar{x}$. When x_0 is not given, we iterate it forward and obtain $x_0 = \beta^{-t}x_t - \sum_{j=1}^t b^{-j}c$, implies $x_0 = \frac{c}{1-b} = \bar{x}$, then $\forall t$, $x_t = \bar{x}$.

In the nonautonomous case, firstly we consider the case in which x_0 and $\{z_t\}_{t=0}^{\infty}$ is exogenously given, and x_t is given by

$$x_t = b^t x_0 + c \sum_{j=0}^{t-1} b^j z_{t-1-j}.$$

- If $|\beta| < 1$, then

$$\lim_{t \rightarrow \infty} x_t = \lim_{t \rightarrow \infty} c \sum_{j=0}^{t-1} b^j z_{t-1-j},$$

where a finite limit exists because we assume $\{z_t\}$ is a bounded sequence. Thus we call this limit a generalized steady state. It is globally asymptotically stable.

- If $|\beta| > 1$, then the solution will diverge.
- If $|\beta| = 1$, then the solution does not converge to a finite limit unless $\sum_{j=0}^{\infty} z_j$ is finite.

Suppose x_0 is not given, we may solve for x_t by forward iteration under the assumption $\lim_{T \rightarrow \infty} b^{-T} x_{t+T} = 0$ which is called transversality condition or no-bubble condition:

$$x_t = b^{-T} x_{t+T} - \frac{c}{b} \sum_{j=0}^{T-1} b^{-j} z_{t+j} = -\frac{c}{b} \sum_{j=0}^{T-1} b^{-j} z_{t+j}.$$

- If $|\beta| > 1$, then the above infinite sum is finite since $\{z_t\}$ is a bounded sequence.
- If $|\beta| < 1$, then there is an infinity of bubble solutions, which are globally stable rather than exploding.

If we remove the transversality condition, then the first-order equation admits many unstable solutions. Let x_t^* denote the solution as above, and for any B_t satisfying

$$B_{t+1} = bB_t,$$

since

$$x_{t+1} = x_{t+1}^* + B_{t+1} = bx_t^* + cz_t + bB_t = bx_t + cz_t,$$

then $x_t = x_t^* + B_t$ is also a solution. We often call the x_t^* fundamental solution and B_t a bubble. The bubble grows at the gross rate b .

2.1.2 Lag Operator

Lag operators provide a powerful tool for solving difference equations. They are also useful for analyzing economic dynamics and time series econometrics.

Definition 2.1.1: Lag Operator

Consider a sequence $\{X_t\}_{t=-\infty}^{+\infty}$, $\forall n$, the lag operator $\mathbf{L}$ on this sequence is defined by

$$\mathbf{L}X_t = X_{t-1}, \quad \mathbf{L}^n X_t = X_{t-n}.$$

In addition, $\mathbf{L}^n c = c$ for any constant c that is independent of time.

Specifically, for a infinity sequence $x_t = [x_1, x_2, \dots]^T$, the lag operator can be written as

$$\mathbf{L} = \begin{bmatrix} 0 & 0 & 0 & 0 & \cdots \\ 1 & 0 & 0 & 0 & \cdots \\ 0 & 1 & 0 & 0 & \cdots \\ 0 & 0 & 1 & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}.$$

The following formulas are useful in applications:

$$\frac{1}{1 - \lambda \mathbf{L}^n} = \sum_{j=0}^{\infty} \lambda^j \mathbf{L}^{nj}, \quad \forall |\lambda| < 1$$

and

$$(I - A \mathbf{L}^n)^{-1} = \sum_{j=0}^{\infty} A^j \mathbf{L}^{nj},$$

if all of the eigenvalues of the matrix A are inside the unit circle.

2.1.3 Second-Order Linear Difference Equations

Consider the following scalar second-order linear difference equation:

$$x_{t+2} = ax_{t+1} + bx_t + cz_t,$$

where $x_0 \in \mathbb{R}$ is exogenously given, a, b, c are real-valued constants, and $\{z_t\}_{t=0}^{\infty}$ is the given bounded sequence.

We write this equation as:

$$(\mathbf{L}^{-2} - a\mathbf{L}^{-1} - b)x_t = cz_t,$$

then factorize the LHS of above equation to derive

$$(\mathbf{L}^{-1} - \lambda_1)(\mathbf{L}^{-1} - \lambda_2)x_t = cz_t,$$

where λ_1, λ_2 are the solution to the characteristic equation,

$$\lambda^2 - a\lambda - b = 0.$$

By the method of factorization ([Sargent, 1987](#)), a particular solution is given by

$$x_t^p = \frac{cz_t}{(\mathbf{L}^{-1} - \lambda_1)(\mathbf{L}^{-1} - \lambda_2)}.$$

If we can find the solution x_t^h to the homogeneous equation $x_{t+2} = ax_{t+1} + bx_t$, i.e., $(\mathbf{L}^{-2} - a\mathbf{L}^{-1} - b)x_t^h = 0$, then the solution is given by $x_t = x_t^p + x_t^h$. Thus, we focus on the homogeneous case below. There are three cases to consider.

Case 1: The two roots λ_1 and λ_2 are real and distinct.

The general solution to the homogeneous equation is given by

$$x_t^h = c_1 \lambda_1^t + c_2 \lambda_2^t,$$

where c_1, c_2 are constants to be determined by the initial value condition and an additional boundary condition. Without loss of generality, we assume $|\lambda_1| < |\lambda_2|$.

- If $|\lambda_1| > 1$ and $|\lambda_2| > 1$, then solution explodes as time proceeds. The steady state is called a source.
- If $|\lambda_1| < 1$ and $|\lambda_2| < 1$, then for any initial value, the solution converges to the steady state. This steady state is called a sink, and the solution is indeterminate.
- If $|\lambda_1| < 1$ and $|\lambda_2| > 1$, then The solution for this case is often called the saddle path solution. The associated steady state is called a saddle point.

In the saddle path case, we can transform the expression into the following form:

$$(\mathbf{L}^{-1} - \lambda_1)x_t = \frac{cz_t}{\mathbf{L}^{-1} - \lambda_2} = \frac{-\frac{c}{\lambda_2}t}{1 - \frac{\mathbf{L}^{-1}}{\lambda_2}}.$$

Since

$$(\mathbf{L}^{-1} - \lambda_1)x_t = x_{t+1} - \lambda_1 x_t,$$

and

$$\frac{-\frac{c}{\lambda_2}t}{1 - \frac{\mathbf{L}^{-1}}{\lambda_2}} = -\frac{c}{\lambda_2} \sum_{j=0}^{\infty} (\lambda^{-1} \mathbf{L}^{-1})^j z_t = -\frac{c}{\lambda_2} \sum_{j=0}^{\infty} \lambda^{-j} z_{t+j},$$

we obtain

$$x_{t+1} = \lambda_1 x_t - \frac{c}{\lambda_2} \sum_{j=0}^{\infty} \lambda^{-j} z_{t+j}.$$

Case 2: The two roots λ_1 and λ_2 are identical.

Denote $\lambda_1 = \lambda_2 = \lambda$, the general solution is given by

$$x_t^h = (c_1 + tc_2)\lambda^t,$$

where c_1 and c_2 are two constants to be determined by two boundary conditions. For stability, we need the two roots to be less than 1. Without imposing a boundary condition at a finite time, the solution is indeterminate since c_1 and c_2 cannot be uniquely determined.

Case 3: The two roots λ_1 and λ_2 are complex.

Suppose that $\lambda_1 = re^{i\theta}$ and $\lambda_2 = re^{-i\theta}$. The general solution is given by

$$x_t^h = c_1 \lambda_1^t + c_2 \lambda_2^t = r^t (c_1 e^{i\theta t} + c_2 e^{-i\theta t}),$$

where c_1 and c_2 are two constants to be determined. If boundary conditions involve real numbers only, then the solution must be real.

We usually impose extra restrictions that only involve the real number, then we have

$$\begin{aligned} x_t^h &= r^t c_1 (\cos(\theta t) + \sin(\theta t)i) + r^t c_2 (\cos(\theta t) - \sin(\theta t)i) \\ &= r^t (c_1 + c_2) \cos(\theta t) + r^t (c_1 - c_2) \sin(\theta t)i. \end{aligned}$$

If c_1 and c_2 are conjugate complex numbers, then both $c_1 + c_2$ and $(c_1 - c_2)i$ are real. Consequently, we can write the solution as

$$x_t^h = r^t [A \cos(\theta t) + B \sin(\theta t)],$$

where A and B are two real numbers to be determined by two boundary conditions.

This solution oscillates with the frequency of $\frac{\theta}{2\pi}$. If $r < 1$, then the oscillations are damped; if $r > 1$, then they are explosive.

2.2 First-Order Linear Systems

An alternative way to solve the second-order linear difference equation is to transform it into a system of linear difference equations by introducing an auxiliary equation $x_{t+1} = x_{t+1}$:

$$\begin{bmatrix} x_{t+1} \\ x_{t+2} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ b & a \end{bmatrix} \begin{bmatrix} x_t \\ x_{t+1} \end{bmatrix} + \begin{bmatrix} 0 \\ c \end{bmatrix} z_t.$$

In this case, x_t is predetermined, but x_{t+1} is non-predetermined. In a similar way, we can transform any higher-order linear difference equation into a system of linear difference equations.

In this section, we consider a general system of the following form:

$$Ax_{t+1} = Bx_t + Cz_t, \quad t = 0, 1, 2, \dots,$$

where $x_t \in \mathbb{R}^n$ is endogenous, $z_t \in \mathbb{R}^{n_z}$ is a vector of exogenous forcing variables, and A , B , C are conformable matrices.

Following [King and Watson \(1998\)](#), we make the following regularity assumption:

$$\exists \alpha \quad \text{s.t.} \quad \det(A\alpha - B) \neq 0.$$

When the pair (A, B) satisfies this assumption, we say it is regular. This assumption is essentially a solvability condition. To see why there may not have a solution if regularity condition is violated, we rewrite the original form as

$$(A\mathbf{L}^{-1} - B)x_t = Cz_t.$$

If $\forall \alpha$, $A\alpha - B$ is singular, then $\exists \psi(\alpha)$ s.t. $\psi(\alpha)(A\alpha - B) = 0$ identically in α , since its rows are linearly dependent. Thus, we have $\psi(\mathbf{L}^{-1})Cz_t = 0$. This equation cannot hold for all sequences $\{z_t\}$.

Prior to the formal discussion, we first introduce the definition of matrix stability.

Definition 2.2.1: Stability

An eigenvalue λ_i of matrix A is stable if $\|\lambda_i\| < 1$, It is unstable if $\|\lambda_i\| > 1$. Matrix A is stable if $\forall i, \|\lambda_i\| < 1$.

2.2.1 The Blanchard and Kahn Method

The solution for the case of nonsingular A is first studied by [Blanchard and Kahn \(1980\)](#). In this case, we rewrite the original form as¹

$$x_{t+1} = A^{-1}Bx_t + A^{-1}Cz_t.$$

Define $W = A^{-1}B$. By the Jordan form, $\exists P$ (left eigenvectors of W) s.t. $W = P^{-1}JP$, where J is a Jordan matrix:

$$J = \begin{bmatrix} J_1 & & & \\ & J_2 & & \\ & & \ddots & \\ & & & J_l \end{bmatrix}, \quad \text{with } J_i = \begin{bmatrix} \lambda_i & 1 & & \\ & \lambda_i & 1 & \\ & & \ddots & 1 \\ & & & \lambda_i \end{bmatrix}_{m_i \times m_i},$$

where $\lambda_1, \dots, \lambda_l$ are the eigenvalues of W satisfying $\det(W - \lambda \mathbf{I}) = 0$.

We can further rewrite the original form as

$$x_{t+1} = P^{-1}JPx_t + A^{-1}Cz_t,$$

then

$$Px_{t+1} = JPx_t + PA^{-1}Cz_t.$$

Define $x_t^* = Px_t$ and $C^* = PA^{-1}C$, then we transform the system into

$$x_{t+1}^* = Jx_t^* + C^*z_t.$$

Solving this equation backward, we obtain the backward-looking solution:

$$x_t^* = J^t x_0^* + \sum_{j=0}^{t-1} J^j C^* z_{t-1-j}, \quad t \geq 1,$$

where

$$J = \begin{bmatrix} J_1^t & & & \\ & J_2^t & & \\ & & \ddots & \\ & & & J_l^t \end{bmatrix}, \quad \text{with } J_i = \begin{bmatrix} \lambda_i^t & t\lambda_i^{t-1} & \binom{t}{2}\lambda_i^{t-2} & \dots \\ & \lambda_i^t & t\lambda_i^{t-1} & \dots \\ & & \ddots & \dots \\ & & & \lambda_i^t \end{bmatrix}.$$

If x_0 is exogenously given, then x_0^* is determined, and the solution only requires of stability of eigenvalues of matrix W . By solving for $\{x_t^*\}$, we obtain the solution $x_t = P^{-1}x_t^*$.

¹If z is not an eigenvalue of $A^{-1}B$, then $\det(Az - B) = \det(A) \det(\mathbf{I}z - A^{-1}B) \neq 0$. Thus the regularity assumption is always satisfied.

If some components of x_0 are not given, in the presence of non-predetermined variables, we cannot use the backward-looking solution. To solve this case, we firstly re-order the components of x_t as $x_t = [k_t, y_t]^T$, where y_t contains n_y non-predetermined variables and k_t contains $n_k = n - n_y$ predetermined variables.

Suppose W has n_u unstable eigenvalues and $n_s = n - n_u$ stable eigenvalues. Assume W has no eigenvalues on the unit circle, which called the hyperbolic condition. we order the eigenvalues from smallest to largest, and partition the Jordan matrix as

$$J = \begin{bmatrix} J_s & \\ & J_u \end{bmatrix},$$

where all eigenvalues of the $n_s \times n_s$ matrix J_s are stable and all eigenvalues of the $n_u \times n_u$ matrix J_u are unstable.

Similarly, we partition x_t^* and C^* as

$$x_t^* = \begin{bmatrix} s_t \\ u_t \end{bmatrix}, \quad C^* = \begin{bmatrix} C_s^* \\ C_u^* \end{bmatrix},$$

so that they are conformable with the partition of J . Then we rewrite the transformed system as

$$\begin{bmatrix} s_{t+1} \\ u_{t+1} \end{bmatrix} = \begin{bmatrix} J_s & \\ & J_u \end{bmatrix} \begin{bmatrix} s_t \\ u_t \end{bmatrix} + \begin{bmatrix} C_s^* \\ C_u^* \end{bmatrix} z_t.$$

The second row of the above system is diverged:

$$u_{t+1} = J_u u_t + C_u^* z_t,$$

which implies

$$(\mathbf{L}^{-1} - J_u)u_t = C_u^* z_t,$$

then

$$\begin{aligned} u_t &= (\mathbf{L}^{-1} - J_u)^{-1} C_u^* z_t \\ &= -J_u^{-1} (\mathbf{I} - \mathbf{L}^{-1} J_u^{-1})^{-1} C_u^* z_t \\ &= -J_u^{-1} \sum_{j=0}^{\infty} (J_u^{-1} \mathbf{L}^{-1})^j C_u^* z_t \\ &= -\sum_{j=0}^{\infty} J_u^{-j-1} C_u^* z_{t+j}, \end{aligned}$$

where we need $\{z_t\}$ to be stable for the series to converge.

Recalling that the transformation of matrix P implies

$$\begin{bmatrix} s_t \\ u_t \end{bmatrix} = P \begin{bmatrix} k_t \\ y_t \end{bmatrix} = \begin{bmatrix} P_{sk} & P_{sy} \\ P_{uk} & P_{uy} \end{bmatrix} \begin{bmatrix} k_t \\ y_t \end{bmatrix},$$

where

$$\begin{bmatrix} P_{sk} & P_{sy} \\ P_{uk} & P_{uy} \end{bmatrix} = \begin{bmatrix} n_s \times n_k & n_s \times n_y \\ n_u \times n_k & n_u \times n_y \end{bmatrix}.$$

The second row implies that

$$u_t = P_{uk}k_t + P_{uy}y_t.$$

Let $t = 0$, we obtain

$$P_{ny}y_0 = u_0 - P_{uk}k_0.$$

Now we give the Blanchard-Kahn conditions to ensure the system has a unique stable solution for any given k_0 and stable sequence $\{z_t\}$:

1. The number of unstable eigenvalues of W is equal to the number of non-predetermined variables, i.e., $n_u = n_y$.
2. P_{uy} is nonsingular.

Under these conditions, since u_0 can be solved as above, y_0 can be represented as

$$y_0 = P_{ny}^{-1}u_0 - P_{ny}^{-1}P_{uk}k_0.$$

Thus, this system can be determined. All points (k_0, y_0) satisfying the above equation form an n_k -dimensional subspace of $\mathbb{R}^n$ is called as the stable manifold.

If the hyperbolic condition is violated, the solution constructed still applies to this case. However, the solution may not be stable. For example, suppose $n_k = 1$, then k_t behaves like a random walk, which is not stable in general.

If the Blanchard-Kahn conditions are violated, then there are two cases.

- If $n_u > n_y$, then there is no solution (a.e.) to the system, since $m(u_0 - P_{uk}k_0) = 0$ in the space of $\mathbb{R}^{n_u} \times \mathbb{R}^{n_y}$.
- If $n_u < n_y$, then there are infinitely many solutions to the system. The solution is indeterminate.

2.2.2 The Klein Method

In many economic problems, the assumption of nonsingular A is violated. In this case, We follow the approach of [Klein \(2000\)](#).

Firstly, we introduce the generalized Schur decomposition which is also called the QZ decomposition.

Definition 2.2.2: Generalized Eigenvalues / Eigenvectors

A scalar $\lambda \in \mathbb{C}$ is a generalized eigenvalue of the matrices (A, B) if there is a nonzero vector $x \in \mathbb{C}^n$ such that $Bx = \lambda Ax$. We call x the corresponding right generalized eigenvector.

Theorem 2.2.3: Complex Generalized Schur Form

If the $n \times n$ matrices A and B are regular, then there exist $n \times n$ unitary matrices of complex numbers Q and Z such that

1. $QAZ = S$ is upper triangular;
2. $QBZ = T$ is upper triangular;
3. $\forall i, s_{ii}, t_{ii} \neq 0$;
4. $\lambda(A, B) = \left\{ \frac{t_{ii}}{s_{ii}} : s_{ii} \neq 0 \right\}$;
5. The pairs (s_{ii}, t_{ii}) . $i = 1, \dots, n$ can be arranged in any order.

Proof. See [Golub and van Loan \(1996\)](#) or any related textbook of matrix computations. $\square$

Let S and T be arranged in such a way that the n_s stable generalized eigenvalues come first. The remaining n_u generalized eigenvalues are unstable. Define $x_t^* = Z^H x_t$ ². We rewrite the original form:

$$QAZZ^H x_{t+1} = QBZZ^H x_t + QCz_t,$$

then

$$Sx_{t+1}^* = Tx_t^* + QZz_t$$

As in the previous subsection, we use the partition:

$$x_t = \begin{bmatrix} k_t \\ y_t \end{bmatrix} \begin{matrix} n_k \times 1 \\ n_y \times 1 \end{matrix}, \quad x_t^* = \begin{bmatrix} s_t \\ u_t \end{bmatrix} \begin{matrix} n_s \times 1 \\ n_u \times 1 \end{matrix}.$$

We also partition Z conformably with the above partition:

$$Z = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{matrix} n_s \times n_k & n_s \times n_y \\ n_u \times n_k & n_u \times n_y \end{matrix},$$

so that

$$\begin{bmatrix} S_{11} & S_{12} \\ 0 & S_{22} \end{bmatrix} \begin{bmatrix} s_{t+1} \\ u_{t+1} \end{bmatrix} = \begin{bmatrix} T_{11} & T_{12} \\ 0 & T_{22} \end{bmatrix} \begin{bmatrix} s_t \\ u_t \end{bmatrix} + \begin{bmatrix} Q_1 \\ Q_2 \end{bmatrix} Cz_t.$$

By the ordering of stable generalized eigenvalues, S_{11} is invertible because all diagonal elements s_{ii} of S_{11} cannot be zero. In addition, T_{22} is also invertible because all diagonal elements t_{ii} of T_{22} cannot be zero.

Similarly to the Blanchard-Kahn condition, We assume $n_s = n_k$ and Z_{11}^{-1} is invertible. The hyperbolic condition also holds.

² $Z^H = Z^{-1}$ since Z is a unitary matrix.

Denote $C_s^* = Q_1 C$ and $C_u^* = Q_2 C$. We can solve $\{u_t\}$ by the second row:

$$\begin{aligned} u_t &= -T_{22}^{-1}(\mathbf{I} - \mathbf{L}^{-1}T_{22}^{-1}S_{22})^{-1}C_u^*z_t \\ &= -T_{22}^{-1}\sum_{j=0}^{\infty}(T_{22}^{-1}S_{22})^jC_u^*z_{t+j}. \end{aligned}$$

In addition, we know that the non-predetermined variable follows

$$k_t = Z_{11}s_t + Z_{12}u_t.$$

Since Z_{11} is invertible by assumption, we obtain

$$s_0 = Z_{11}^{-1}(k_0 - Z_{12}u_0).$$

By the first row we have

$$S_{11}s_{t+1} + S_{12}u_{t+1} = T_{11}s_t + T_{12}u_t + C_s^*z_t,$$

then manipulating it yields the backward-looking solution:

$$s_{t+1} = S_{11}^{-1}(T_{11}s_t + T_{12}u_t - S_{12}u_{t+1} + C_s^*z_t).$$

For given s_0 and $\{u_t\}$, we can solve $\{s_t\}$ as above. Thus, $\{k_t\}$ and $\{y_t\}$ are also solved.

2.3 Nonlinear Systems

Consider the following first-order nonlinear system:

$$f(x_{t+1}, x_t, z_t) = 0,$$

where $x_t \in \mathbb{R}^n$ is endogenous, $z_t \in \mathbb{R}^{K_F}$ is a vector of exogenous forcing variables, and $f : \mathbb{R}^{2n} \times \mathbb{R}^{K_F} \rightarrow \mathbb{R}^n$. Many high-order nonlinear systems can be transformed to a first-order system by a suitable change of variables as we have done previously.

We assume the steady state $\bar{x}$ exists, then take the first-order Taylor approximation about $\bar{x}$:

$$0 = f_{x'}x_{t+1} + f_x x_t + f_z z_t,$$

where $f_{x'} = \frac{\partial f}{\partial x_{t+1}}$, $f_x = \frac{\partial f}{\partial x_t}$, and $f_z = \frac{\partial f}{\partial z_t}$, respectively, evaluated at the steady state.

Since the equation above is a linear system, we may apply the method discussed before to analyze this system. The critical conditions for stability are based on the properties of the generalized eigenvalues of the pair of Jacobian $(f_{x'}, -f_x)$ evaluated at the steady state.

Chapter 3

Dynamic Programming

3.1 Markov Decision Process Model

Economic decisions can often be summarized as a problem in which a decision maker chooses an action to influence a stochastic dynamic system so as to maximize some criterion.

To ensure a recursive structure, we formulate the framework by a Markov decision process, which consists of five elements:

- Time: The decisions are made at discrete points in time, denoted by $t \in \mathcal{T} \equiv \{0, 1, \dots, T\}$, where $T \leq \infty$.
- States: At each time t , the system occupies a state s_t . The initial state s_0 is exogenously given. Denote the set of all possible state by $\mathbb{S}$, call the state space.
- Actions: Denote $\mathbb{A}$ as the action space which consists of all possible action¹. After observing the state s_t , the decision maker choose an action $a_t \in \mathbb{A}$. Moreover, The constraints on this action are described by a correspondence $\Gamma : \mathbb{S} \rightarrow \mathbb{A}$, that is, $\Gamma(s_t) \subset \mathbb{A}$ is the set of feasible actions given that the current state is $s_t \in \mathbb{S}$.
- Rewards: As a result of choosing action a_t at state s_t at time t , the decision maker receives a one-period reward $u_t(s_t, a_t)$.
- Stochastic Kernel: The system state at time $t + 1$ is determined by the stochastic kernel $P_{t+1} : (\mathbb{S} \times \mathbb{A}) \times \mathcal{S} \rightarrow [0, 1]$, that is, $P_{t+1}(s_t, a_t; \cdot)$ is the probability measure for the set of states at time $t + 1$, if the action and state at time t are s_t and a_t , respectively². In addition, $\forall B \in \mathcal{S}$, $P_{t+1}(\cdot, \cdot; B)$ is the measure function on $\mathbb{S} \times \mathbb{A}$. Define the Markov semigroup as a group of stochastic kernel $\mathcal{P} = \{P_t\}_{t=1}^T$.

We can define the Markov decision process as follows:

$$\{\mathcal{T}, (\mathbb{S}, \mathcal{S}), (\mathbb{A}, \mathcal{A}), \Gamma, \mathcal{P}, \{u_t\}_{t=0}^T\}.$$

¹Let $(\mathbb{S}, \mathcal{S})$ and $(\mathbb{A}, \mathcal{A})$ be some measurable spaces, where $\mathcal{S}$ ($\mathcal{A}$) is the σ -algebra of $\mathbb{S}$ ($\mathbb{A}$).

²If the transition function is $s_{t+1} = f(s_t, a_t)$, then the probability measure becomes $P_{t+1}(s_t, a_t; B) = \mathbb{I}(f(s_t, a_t) \in B)$.

We have assumed that S , A and Γ are independent of time, but P_{t+1} and u_t may depend on time. For the infinite-horizon case, we shall assume that P_{t+1} and u_t are independent of time. In this case, the Markov decision process is stationary.

In sum, a Markov decision process has the following essential features:

- The state system $\{s_t\}_{t=0}^T$ is governed by a Markov process.
- The stochastic kernels $\mathcal{P}$ of this Markov process are affected by the decision maker's current actions, independent of future or past actions.
- The constraints correspondence Γ maps from a current state to a set of actions, independent of histories.
- The one-period rewards u_t are a function of current states and current actions, independent of histories. In the finite-horizon case, the terminal reward is either zero or a function of state only.

A Markov decision process together with an optimality criterion is called a Markov decision problem. A Markov decision process and the associated decision problem form a Markov decision process model. We will give the formal definition of the Markov decision problem. Define the product measurable spaces $(\mathbb{S}^t, \mathcal{S}^t)$, $t = 1, \dots, T$.

Definition 3.1.1: Plan

A plan or policy π is a sequence of actions $(\pi_0, \pi_1, \dots)$ such that $\pi_0 \in \mathbb{A}$ and $\pi_t : \mathbb{S}^t \rightarrow \mathbb{A}$ is measurable. It is Markovian if π_t is independent of $\{s_t\}_{t=0}^{t-1}$. It is stationary if π_t is independent of time t . It is feasible from $s_0 \in \mathbb{S}$ if $\pi_0 \in \Gamma(s_0)$ and $\forall t, \forall s^t \in \mathbb{S}^t$, $\pi_t(s^t) \in \Gamma(s_t)$.

We next define the expectation with respect to decision maker's states and action. Given an initial state s_0 , a feasible plan π from s_0 , and the stochastic kernels $\mathcal{P}$, define a probability measure on $\mathcal{S}^t$ as:

$$\mu^t(s_0, B; \pi) = \int_{B_1} \cdots \int_{B_{t-1}} \int_{B_t} \prod_{i=1}^t P_i(s_{i-1}, \pi_{i-1}(s^{i-1}); ds_i) \times P_1(s_0, \pi_0; ds_1),$$

where $B = \prod_{i=1}^t B_i \in \mathcal{S}^t$. In the infinite-horizon case, the sequence of measures $\mu^t(s_0, \cdot; \pi)$, $t = 1, 2, \dots$, induces a unique measure $\mu^\infty(s_0, \cdot; \pi)$ on $\mathcal{S}^\infty$ by the Kolmogorov Extension Theorem.

For any plan $\pi \in \Pi(s_0)$, where $\Pi(s_0)$ denote the set of all feasible plans from s_0 , the total discounted expected rewards for $T < \infty$ is defined as:

$$J_0^T(s_0, \pi) = u_0(s_0, \pi) + \sum_{t=1}^{T-1} \beta^t \int u_t(s_t, \pi_t(s^t)) \mu^t(s_0, ds^t; \pi) + \beta^T \int u_T(s_T) \mu^T(s_0, ds^T; \pi).$$

If $T = \infty$, we define the total discounted expected rewards as:

$$J(s_0, \pi) = u_0(s_0, \pi) + \lim_{T \rightarrow \infty} \sum_{t=1}^T \beta^t \int u_t(s_t, \pi_t(s^t)) \mu^t(s_0, ds^t; \pi).$$

We assume that the above reward function is integrable and the limit exists. Then, for each $s \in \mathbb{S}$, if $T < \infty$ the optimality criterion is defined as:

$$V_0(s) = \sup_{\pi \in \Pi(s)} J_0^T(s, \pi),$$

or if $T = \infty$:

$$V(s) = \sup_{\pi \in \Pi(s)} J(s, \pi).$$

In an infinite-horizon Markov decision problem, $V : \mathbb{S} \rightarrow \bar{\mathbb{R}}$ is called the value function.

3.2 Infinite-Horizon Dynamic Programming

In this section, we will prove the infinite-horizon Principle of Optimality by establishing three theorems. Recall that the value function is defined by

$$V(s) = \sup_{\pi \in \Pi(s)} J(s, \pi),$$

where $\Pi(s)$ is the set of feasible plans from initial state $s \in \mathbb{S}$, and $J(s, \pi)$ is the total discounted expected rewards from π .

Lemma 3.2.1

Let $(\mathbb{S}, \mathcal{S})$, $(\mathbb{A}, \mathcal{A})$, Γ , P , u , and β be given, and a series of assumptions satisfied, then for any $s_0 \in \mathbb{S}$ and any $\pi \in \Pi(s_0)$,

$$J(s_0, \pi) = u(s_0, \pi_0) + \beta \int J(s_1, C(\pi, s_1)) P(s_0, \pi_0; ds_1),$$

where $C(\pi, s_1)$ is the continuation of plan π following s_1 .

Proof. By definition,

$$J(s_0, \pi) = u(s_0, \pi_0) + \beta \lim_{n \rightarrow \infty} \sum_{t=1}^n \beta^{t-1} \int u(s_t, \pi_t(s^t)) \mu^t(s_0, ds^t; \pi).$$

The second term on the right-hand side can be transformed as:

$$\begin{aligned} & \lim_{n \rightarrow \infty} \sum_{t=1}^n \beta^{t-1} \int u(s_t, \pi_t(s^t)) \mu^t(s_0, ds^t; \pi) \\ &= \lim_{n \rightarrow \infty} \int \left[u(s_1, \pi_1(s^1)) + \sum_{t=2}^n \beta^{t-1} \int u_t(s_t, \pi_t(s^t)) \mu^{t-1}(s_1, ds_2^t; C(\pi, s_1)) \right] P(s_0, \pi_0; ds_1) \\ &= \int \lim_{n \rightarrow \infty} u(s_1, \pi_1(s^1)) + \sum_{t=2}^n \beta^{t-1} \int u_t(s_t, \pi_t(s^t)) \mu^{t-1}(s_1, ds_2^t; C(\pi, s_1)) P(s_0, \pi_0; ds_1) \\ &= \int J(s_1, C(\pi, s_1)) P(s_0, \pi_0; ds_1), \end{aligned}$$

where the second equality from the Monotone Convergence Theorem if $u \geq 0$ or $u \leq 0$, and from the Dominated Convergence Theorem if u take on both signs. $\square$

Theorem 3.2.2: Dynamic Programming Principle

Let $(\mathbb{S}, \mathcal{S})$, $(\mathbb{A}, \mathcal{A})$, Γ , P , u , and β be given, and a series of assumptions satisfied, then the value function $V(s)$ satisfies the following Bellman equation:

$$V(s) = \sup_{a \in \Gamma(s)} u(s, a) + \int V(s') P(s, a; ds')$$

Proof. For any plan $\pi \in \Pi(s)$, by the property of supremum, we have

$$\begin{aligned} V^*(s) &\equiv \sup_{a \in \Gamma(s)} u(s, a) + \int V(s') P(s, a; ds') \\ &\geq \sup_{a \in \Gamma(s)} u(s, a) + \int J(s', C(s', \pi)) P(s, a; ds') \\ &\geq u(s, \pi_0) + \int J(s', C(s', \pi)) P(s, a; ds') \\ &= V(s). \end{aligned}$$

Thus, $V^*(s) \geq V(s)$.

By assumption, $\forall \varepsilon > 0$, $\exists s, \pi^s \in \Pi(s)$ s.t. $J(s, \pi) + \varepsilon \geq V(s)$. Thus, by Lemma 2.2.1, we have,

$$\begin{aligned} V(s) &= \sup_{\pi \in \Pi(s)} u(s, \pi_0) + \beta \int (s', C(s', \pi)) P(s, a; ds') \\ &\geq \sup_{a \in \Gamma(s)} u(s, a) + \beta \int (V(s') - \varepsilon) P(s, a; ds') \\ &= V^*(s) - \beta \varepsilon. \end{aligned}$$

Let $\varepsilon \rightarrow 0$, we obtain $V(s) \geq V^*(s)$. Combining the above two steps completes the proof. $\square$

This theorem demonstrates that the value function implied from sequential problem is equivalent to that of Bellman equation. Moreover, in dynamic programming problem, if there is a solution $a \in \Gamma(s)$ that attains the supremum in Bellman equation, we denote the set of all solutions as $G(s)$. This defines an optimal policy correspondence $G : \mathbb{S} \rightarrow \mathbb{A}$.

The following result demonstrates that under some conditions the solution to the Bellman equation gives the value function for the Markov decision problem. In addition, any plan generated by the optimal policy correspondence from the dynamic programming problem is optimal for the Markov decision problem.

Theorem 3.2.3: Verification

Let $(\mathbb{S}, \mathcal{S})$, $(\mathbb{A}, \mathcal{A})$, Γ , P , u , and β be given, and a series of assumptions satisfied. Suppose that $V^* : \mathbb{S} \rightarrow \mathbb{R}$ is a measurable function satisfying the Bellman equation, and $\forall \pi \in \Pi(s_0)$, $\forall s_0 \in \mathbb{S}$,

$$\lim_{n \rightarrow \infty} \beta^n \int V^*(s_n) \mu^n(s_0, ds^n; \pi) = 0.$$

Let G be the optimal policy correspondence associated with the Bellman equation, and suppose that G is nonempty and permits a measurable selection. Then $V = V^*$ and any plan π^* generated by G also attains the supremum of Markov decision problem.

Proof. $\forall \pi \in \Pi(s_0)$, $\forall s_0 \in \mathbb{S}$, by the property of supremum in Bellman equation, we derive:

$$V^*(s_t) \geq u(s_t, \pi_t(s^t)) + \beta \int V^*(s_{t+1}) P(s_t, \pi_t(s^t); ds_{t+1}),$$

for $t = 0, 1, \dots, n$. Multiplying by β^t and rearranging yield:

$$\beta^t V^*(s_t) - \beta^{t+1} \int V^*(s_{t+1}) P(s_t, \pi_t(s^t); ds_{t+1}) \geq \beta^t u(s_t, \pi_t(s^t)).$$

Taking expectations with $\mu^t(s_0, ds^t; \pi)$ and summing, we obtain:

$$V^*(s_0) - \beta^n \int V^*(s_n) \mu^n(s_0, ds^n; \pi) \geq \sum_{t=0}^{n-1} \beta^t u(s_t, \pi_t(s^t)) \mu^t(s_0, ds^t; \pi).$$

Taking limit as $n \rightarrow \infty$ and using the so-called transversality condition, we deduce

$$V^*(s_0) \geq J(s_0, \pi).$$

Thus, $\forall \pi \in \Pi(s_0)$, $V^*(s_0) \geq V(s_0)$. The equality is attained if we replace π by π^* since it is generated by the optimal policy correspondence G , so does $\forall t$ at $\pi = \pi^*$. $\square$

The final result is a partial converse of Theorem 2.2.3. It ensures that any optimal plan for the Markov decision problem can be found by solving the dynamic programming problem.

Theorem 3.2.4

Let $(\mathbb{S}, \mathcal{S})$, $(\mathbb{A}, \mathcal{A})$, Γ , P , u , and β be given, and a series of assumptions satisfied. Suppose that $V^* : \mathbb{S} \rightarrow \mathbb{R}$ is a measurable function satisfying the Bellman equation. Suppose that the associated optimal policy correspondence G is nonempty and permits a measurable selection. Then for any optimal plan $\pi^* \in \Pi(s_0)$ for Markov decision problem, $\forall t$, s^t , $\pi_0^* \in G(s_0)$ and $\pi_t^*(s^t) \in G(s_t)$.

Proof. See [Miao \(2020\)](#) due to the space limitation. $\square$

Chapter 4

New Keynesian Models

4.1 Basic New Keynesian Model

4.1.1 Households

Assume a representative infinitely-lived household, seeking to maximize

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U(C_t, N_t), \quad (4.1.1)$$

where $\beta \in (0, 1)$ denotes the discount factor, N_t denotes hours of work or employment, and C_t is a consumption index given by

$$C_t \equiv \left(\int_0^1 C_t(i)^{\frac{\varepsilon-1}{\varepsilon}} di \right)^{\frac{\varepsilon}{\varepsilon-1}},$$

with $C_t(i)$ representing the quantity of good i consumed by the household in period t . Assume the existence of a continuum of goods represented by the interval $[0, 1]$ ¹.

The period budget constraint now takes the form

$$\int_0^1 P_t(i) C_t(i) di + Q_t B_t \leq B_{t-1} + W_t N_t + T_t, \quad (4.1.2)$$

where W_t is the nominal wage, B_t represents purchases of one-period bonds with the price $Q_t = \frac{1}{1+I_t}$, I_t is the nominal interest, and T_t is a lump-sum component of income.

Additionally, assumed that the household is subject to a solvency constraint that prevents it from engaging in Ponzi-type schemes, by the following constraint:

$$\forall t, \lim_{T \rightarrow \infty} \mathbb{E}_t(B_T) \geq 0.$$

Define Z_t as the total nominal expenditure, $P_t(i)$ as the price of good i . The optimal

¹Assume that $U_C > 0$, $U_{CC} \leq 0$, $U_N \leq 0$, and $U_{NN} \leq 0$.

allocation of any given expenditure level can be found by solving

$$\max_{C_t(i)} C_t \equiv \left(\int_0^1 C_t(i)^{\frac{\varepsilon-1}{\varepsilon}} di \right)^{\frac{\varepsilon}{\varepsilon-1}},$$

s.t.

$$\int_0^1 P_t(i) C_t(i) di \equiv Z_t.$$

Lagrangian:

$$\mathcal{L} = \left(\int_0^1 C_t(i)^{\frac{\varepsilon-1}{\varepsilon}} di \right)^{\frac{\varepsilon}{\varepsilon-1}} - \lambda_t \left(\int_0^1 P_t(i) C_t(i) di - Z_t \right).$$

FOC:

$$\frac{\partial \mathcal{L}}{\partial C_t(i)} : C_t^{\frac{1}{\varepsilon}} C_t(i)^{-\frac{1}{\varepsilon}} = \lambda_t P_t(i), \quad \forall i \in [0, 1],$$

that is, $\forall i, j \in [0, 1]$,

$$C_t(i) = C_t(j) \left[\frac{P_t(i)}{P_t(j)} \right]^{-\varepsilon}.$$

Then the isoelastic demand of good i can be represented as

$$C_t(i) = \left[\frac{P_t(i)}{P_t} \right]^{-\varepsilon} C_t,$$

where P_t denote the aggregate price index

$$P_t \equiv \left[\int_0^1 P_t(i)^{1-\varepsilon} di \right]^{\frac{1}{1-\varepsilon}}.$$

Furthermore, we have

$$\int_0^1 P_t(i) C_t(i) di = P_t C_t, \quad (4.1.3)$$

i.e., total consumption expenditures can be written as the product of the price index times the quantity index.

Next we consider the dynamic problem. Combining 4.1.1, 4.1.2 and 4.1.3, the intertemporal UMP can be written as

$$\max_{C_t, N_t, B_t} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U(C_t, N_t),$$

s.t.

$$P_t C_t + Q_t B_t \leq B_{t-1} + W_t N_t + T_t.$$

Current-value Hamiltonian:

$$\mathcal{H} = U(C_t, N_t) - \lambda_t (P_t C_t + Q_t B_t - B_{t-1} - W_t N_t - T_t).$$

FOCs:

$$\frac{\partial \mathcal{H}}{\partial C_t} : U_{C,t} = \lambda_t P_t, \quad (4.1.4)$$

$$\frac{\partial \mathcal{H}}{\partial N_t} : -U_{N,t} = \lambda_t W_t, \quad (4.1.5)$$

$$\frac{\partial \mathcal{H}}{\partial B_t} : \lambda_t Q_t = \beta \mathbb{E}_t \lambda_{t+1}. \quad (4.1.6)$$

Combining 4.1.4 and 4.1.5, we obtain the optimal labor supply decision:

$$-\frac{U_{N,t}}{U_{C,t}} = \frac{W_t}{P_t}.$$

Combining 4.1.4 and 4.1.6, we obtain the Euler equation:

$$Q_t = \beta \mathbb{E}_t \left[\frac{U_{C,t+1}}{U_{C,t}} \frac{P_t}{P_{t+1}} \right].$$

Under the assumption of a period utility given by

$$U(C_t, N_t) = \frac{C_t^{1-\sigma}}{1-\sigma} - \frac{N_t^{1+\varphi}}{1+\varphi},$$

where $\sigma > 0$, $\sigma \neq 1$, and $\varphi > 0$, then the two equations become

$$C_t^\sigma N_t^\varphi = \frac{W_t}{P_t}, \quad (4.1.7)$$

$$Q_t = \beta \mathbb{E}_t \left[\frac{C_{t+1}^{-\sigma}}{C_t^{-\sigma}} \frac{P_t}{P_{t+1}} \right]. \quad (4.1.8)$$

Log-linearize 4.1.7, we obtain the log-linearized optimal labor supply decision:

$$\sigma \hat{c}_t + \varphi \hat{n}_t = \hat{w}_t - \hat{p}_t. \quad (4.1.9)$$

This shows that, given the marginal utility of consumption (which under the assumptions is a function of consumption only), this equation determines labor supply as a function of the real wage.

Log-linearize 4.1.8, we have

$$\hat{q}_t = -\sigma \mathbb{E}_t \hat{c}_{t+1} + \sigma \hat{c}_t + \hat{p}_t - \mathbb{E}_t \hat{p}_{t+1}.$$

Denote $\hat{\pi}_t = \hat{p}_t - \hat{p}_{t-1}$ as the (log-linearized) inflation. Since $\hat{i}_t = -\hat{q}_t$, we obtain:

$$\hat{c}_t = \mathbb{E}_t \hat{c}_{t+1} - \sigma^{-1} (\hat{i}_t - \mathbb{E}_t \hat{\pi}_{t+1}). \quad (4.1.10)$$

This shows that, given the nominal interest, the intertemporal consumption decision is only determined by the expected inflation.

4.1.2 Firms

Assume a continuum of firms indexed by $i \in [0, 1]$. Each firm produces a differentiated good, but they all use an identical technology, represented by the production function

$$Y_t(i) = A_t N_t(i)^{1-\alpha},$$

where $\alpha \in (0, 1)$, A_t represents the level of technology, assumed to be common to all firms and to evolve exogenously over time. All firms face an identical isoelastic demand schedule, and take the aggregate price level P_t and aggregate consumption index C_t as given.

Consider the optimal labor demand for firms. One minimizes its cost by choosing $N_t(i)$, that is,

$$\min_{N_t(i)} \frac{W_t}{P_t} N_t(i),$$

s.t.

$$Y_t(i) = A_t N_t(i)^{1-\alpha}.$$

Lagrangian:

$$\mathcal{L} = \frac{W_t}{P_t} N_t(i) - \lambda_t [A_t N_t(i)^{1-\alpha} - Y_t(i)].$$

FOC:

$$\frac{\partial \mathcal{L}}{\partial N_t(i)} : W_t = (1 - \alpha) \lambda_t A_t N_t(i)^{-\alpha}.$$

Denote $MC_t = \lambda_t P_t$ as the nominal marginal cost. Then the FOC becomes

$$W_t = MC_t (1 - \alpha) A_t N_t(i)^{-\alpha}.$$

In flexible price system, the firm's PMP can be represented as

$$\max_{P_t(i)} P_t(i) Y_t(i) - W_t(i) N_t(i).$$

Dividing P_t in this form. By the optimal labor demand decision, when good market is clear, the objective function can be transformed as

$$\begin{aligned} & \frac{P_t(i) Y_t(i) - W_t(i) N_t(i)}{P_t} \\ &= \left[\frac{P_t(i)}{P_t} - (1 - \alpha) \frac{MC_t}{P_t} \right] C_t(i) \\ &= \left[\left[\frac{P_t(i)}{P_t} \right]^{1-\varepsilon} - (1 - \alpha) \frac{MC_t}{P_t} \left[\frac{P_t(i)}{P_t} \right]^{-\varepsilon} \right] C_t. \end{aligned}$$

FOC:

$$(1 - \varepsilon) \left[\frac{P_t(i)}{P_t} \right]^{-\varepsilon} + (1 - \alpha) \varepsilon \frac{MC_t}{P_t} \left[\frac{P_t(i)}{P_t} \right]^{-\varepsilon-1} = 0.$$

Thus, the optimal price becomes

$$P_t^*(i) = \frac{1 - \alpha}{1 - \varepsilon} MC_t.$$

In this form, the (nominal) markup $\frac{1-\alpha}{1-\varepsilon-1}$ leads to inefficiently low level of employment and output in the economy due to the monopolistic competition.

Then we turn to sticky price system. Following [Calvo \(1983\)](#), each firm may reset its price only with probability $1 - \theta$ in any given period, independent of the time elapsed since the last adjustment, i.e., follows an exogenous Poisson process. Thus, each period a measure $1 - \theta$ of producers reset their prices, while a fraction θ keep their prices unchanged. As a result, the average duration of a price is given by $(1 - \theta)^{-1}$. In this context, θ becomes a natural index of price stickiness.

Let $S(t) \subset [0, 1]$ represent the set of firms not re-optimizing their posted price in period t . The aggregate price index can be written as

$$\begin{aligned} P_t &= \left[\int_{S(t)} P_{t-1}(i)^{1-\varepsilon} + (1 - \theta)(P_t^*)^{1-\varepsilon} \right]^{\frac{1}{1-\varepsilon}} \\ &= [\theta P_{t-1}^{1-\varepsilon} + (1 - \theta)(P_t^*)^{1-\varepsilon}]^{\frac{1}{1-\varepsilon}}. \end{aligned}$$

Dividing P_{t-1} on both sides and rearranging, we obtain

$$\Pi_t^{1-\varepsilon} = \theta + (1 - \theta) \left(\frac{P_t^*}{P_{t-1}} \right)^{1-\varepsilon},$$

where $\Pi_t \equiv \frac{P_t}{P_{t-1}}$ is the gross inflation rate between $t-1$ and t .

The log-linearized form:

$$\hat{p}_t = \theta \hat{p}_{t-1} + (1 - \theta) \hat{p}_t^*, \quad (4.1.11)$$

or

$$\hat{\pi}_t = (1 - \theta)(\hat{p}_t^* - \hat{p}_{t-1}).$$

This implies that, inflation results from the fact that firms re-optimizing in any given period choose a price that differs from the economy's average price in the previous period.

With sticky price, the firm must set its price taking account of the risk that it will not be allowed to change its price in the future. That is, firm choose P_t^* to maximize

$$\begin{aligned} &[P_t^*(i)Y_t(i) - \Psi_t(i)] \\ &+ \theta \mathbb{E}_t[Q_{t,t+1}[P_t^*(i)Y_{t+1|t}(i) - \Psi_{t+1}(i)]] \\ &+ \theta^2 \mathbb{E}_t[Q_{t,t+2}[P_t^*(i)Y_{t+2|t}(i) - \Psi_{t+2}(i)]] \\ &+ \dots, \end{aligned}$$

where $Q_{t,t+k} \equiv \beta^k \left(\frac{C_{t+k}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+k}}$ is the stochastic discount factor for nominal payoffs, $\Psi(\cdot)$ is the cost function, and $Y_{t+k|t}$ denotes output in period $t+k$ for a firm that last reset its price in period t .

Formally, the PMP can be represented as

$$\max_{P_t^*} \sum_{k=0}^{\infty} \theta^k \mathbb{E}_t[Q_{t,t+k}[P_t^*Y_{t+k|t} - \Psi_{t+k}(Y_{t+k|t})]],$$

s.t.

$$Y_{t+k|t} = \left(\frac{P_t^*}{P_{t+k}} \right)^{-\varepsilon} Y_{t+k}, \quad k = 0, 1, 2, \dots$$

The log-linearized form of constraint:

$$\hat{y}_{t+k} - \hat{y}_{t+k|t} = \varepsilon(\hat{p}_t^* - \hat{p}_{t+k}) \quad (4.1.12)$$

FOC:

$$\sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left[Q_{t,t+k} \left(Y_{t+k|t} + P_t^* \frac{\partial Y_{t+k|t}}{\partial P_t^*} - \frac{\partial \Psi_{t+k}}{\partial Y_{t+k|t}} \frac{\partial Y_{t+k|t}}{\partial P_t^*} \right) \right] = 0,$$

where

$$\frac{\partial Y_{t+k|t}}{\partial P_t^*} = -\varepsilon \frac{Y_{t+k|t}}{P_t^*}, \quad k = 0, 1, 2, \dots$$

Denote nominal marginal cost as $\psi_{t+k|t} \equiv \frac{\partial \Psi_{t+k}}{\partial Y_{t+k|t}}$, then

$$\begin{aligned} & \sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left[Q_{t,t+k} \left(Y_{t+k|t} + P_t^* \frac{\partial Y_{t+k|t}}{\partial P_t^*} - \frac{\partial \Psi_{t+k}}{\partial Y_{t+k|t}} \frac{\partial Y_{t+k|t}}{\partial P_t^*} \right) \right] \\ &= \sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left[Q_{t,t+k} \left(Y_{t+k|t} + (P_t^* - \psi_{t+k|t})(-\varepsilon) \frac{Y_{t+k|t}}{P_t^*} \right) \right] \\ &= \sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left[Q_{t,t+k} Y_{t+k|t} \left[1 + (P_t^* - \psi_{t+k|t}) \frac{-\varepsilon}{P_t^*} \right] \right] \\ &= \sum_{k=0}^{\infty} \theta^k \mathbb{E}_t \left[Q_{t,t+k} Y_{t+k|t} \left(1 - \varepsilon + \varepsilon \frac{\psi_{t+k|t}}{P_t^*} \right) \right] = 0. \end{aligned}$$

Solving for the optimal price:

$$P_t^* = \mu \frac{\sum_{k=0}^{\infty} \theta^k \mathbb{E}_t (Q_{t,t+k} Y_{t+k|t} \psi_{t+k|t})}{\sum_{k=0}^{\infty} \theta^k \mathbb{E}_t (Q_{t,t+k} Y_{t+k|t})}, \quad (4.1.13)$$

where $\mu \equiv \frac{\varepsilon}{\varepsilon-1}$.

In the case of flexible price, i.e., $\theta = 0$, 4.1.11 becomes $P_t^* = \mu \psi_{t|t}$. Henceforth, μ is referred to as the desired or frictionless markup.

Denote $MC_t^r = \frac{\psi_t}{P_t}$ as the real marginal cost. Consider the log-linearized form of 4.1.13:

$$\hat{p}_t^* = (1 - \beta\theta) \sum_{k=0}^{\infty} (\beta\theta)^k \mathbb{E}_t (\hat{m}c_{t+k|t}^r + \hat{p}_{t+k}). \quad (4.1.14)$$

The forward-looking equation:

$$\hat{p}_t^* = (1 - \beta\theta) (\hat{m}c_{t|t}^r + \hat{p}_t) + \beta\theta \mathbb{E}_t \hat{p}_{t+1}^*. \quad (4.1.15)$$

Substituting 4.1.15 into 4.1.11, we obtain

$$\begin{aligned}
\hat{p}_t &= \theta \hat{p}_{t-1} + (1 - \theta) \hat{p}_t^* \\
&= \theta \hat{p}_{t-1} + (1 - \theta)(1 - \beta\theta)(\hat{m}c_{t|t}^r + \hat{p}_t) + (1 - \theta)\beta\theta \mathbb{E}_t \hat{p}_{t+1}^* \\
&= \theta \hat{p}_{t-1} + (1 - \theta)(1 - \beta\theta)(\hat{m}c_{t|t}^r + \hat{p}_t) + (1 - \theta)\beta\theta \frac{\mathbb{E}_t \hat{p}_{t+1} - \theta \hat{p}_t}{1 - \theta} \\
&= \theta \hat{p}_{t-1} + (1 - \theta)(1 - \beta\theta)(\hat{m}c_{t|t}^r + \hat{p}_t) + \beta\theta \mathbb{E}_t \hat{p}_{t+1} - \beta\theta^2 \hat{p}_t.
\end{aligned}$$

Subtracting by $(1 - \theta)\hat{p}_t$ and rearranging yields

$$\begin{aligned}
\theta \hat{\pi}_t &\equiv \theta(\hat{p}_t - \hat{p}_{t-1}) \\
&= (1 - \theta)(1 - \beta\theta)\hat{m}c_{t|t}^r + \beta\theta(\mathbb{E}_t \hat{p}_{t+1} - \hat{p}_t) \\
&= (1 - \theta)(1 - \beta\theta)\hat{m}c_{t|t}^r + \beta\theta \mathbb{E}_t \hat{\pi}_{t+1}.
\end{aligned}$$

The forward-looking equation for inflation:

$$\hat{\pi}_t = \beta \mathbb{E}_t \hat{\pi}_{t+1} + \frac{(1 - \theta)(1 - \beta\theta)}{\theta} \hat{m}c_{t|t}^r.$$

4.1.3 Non-Policy Equilibrium

Market clearing in the goods market requires

$$Y_t(i) = C_t(i), \quad \forall i \in [0, 1].$$

Letting aggregate output be defined as $Y_t \equiv \left(\int_0^1 Y_t(i)^{\frac{\varepsilon-1}{\varepsilon}} di \right)^{\frac{\varepsilon}{\varepsilon-1}}$ it follows that $Y_t = C_t$, i.e., $\hat{y}_t = \hat{c}_t$. Combining with 4.1.10, we obtain the equilibrium condition of good market clearing:

$$\hat{y}_t = \mathbb{E}_t \hat{y}_{t+1} - \sigma^{-1}(\hat{i}_t - \mathbb{E}_t \hat{\pi}_{t+1}). \quad (4.1.16)$$

Market clearing in the labor market requires

$$N_t = \int_0^1 N_t(i) di.$$

Combining with the production function, we obtain

$$\begin{aligned}
N_t &= \int_0^1 \left[\frac{Y_t(i)}{A_t} \right]^{\frac{1}{1-\alpha}} di \\
&= \left(\frac{Y_t}{A_t} \right)^{\frac{1}{1-\alpha}} \int_0^1 \left[\frac{P_t(i)}{P_t} \right]^{-\frac{\varepsilon}{1-\alpha}} di.
\end{aligned}$$

Taking the square $1 - \alpha$ yields

$$N_t^{1-\alpha} = \frac{Y_t}{A_t} \left[\int_0^1 \left[\frac{P_t(i)}{P_t} \right]^{-\frac{\varepsilon}{1-\alpha}} di \right]^{1-\alpha}.$$

Denote $D_t = \left[\int_0^1 \left[\frac{P_t(i)}{P_t} \right]^{-\frac{\varepsilon}{1-\alpha}} di \right]^{1-\alpha}$ measuring price (output) dispersion across firms,

we have

$$Y_t = A_t N_t^{1-\alpha} D_t^{-1}.$$

The log-linearized form²:

$$\hat{y}_t = \hat{a}_t + (1 - \alpha)\hat{n}_t. \quad (4.1.17)$$

Now we consider the average real marginal cost in the equilibrium. In firm's CMP, we have known that

$$MC_t^r = \frac{W_t}{P_t} \frac{1}{A_t(1 - \alpha)N_t^{-\alpha}}.$$

The log-linearized form:

$$\begin{aligned} \hat{m}c_t^r &= \hat{w}_t - \hat{p}_t - (\hat{a}_t - \alpha\hat{n}_t) \\ &= \hat{w}_t - \hat{p}_t - \frac{1}{1 - \alpha}(\hat{a}_t - \alpha\hat{y}_t). \end{aligned}$$

For firms that set price at t and remains it at $t + k$:

$$\begin{aligned} \hat{m}c_{t+k|t}^r &= \hat{w}_{t+k} - \hat{p}_{t+k} - \frac{1}{1 - \alpha}(\hat{a}_{t+k} - \alpha\hat{y}_{t+k|t}) \\ &= \hat{w}_{t+k} - \hat{p}_{t+k} - \frac{1}{1 - \alpha}(\hat{a}_{t+k} - \alpha\hat{y}_{t+k|t}) \\ &= \hat{w}_{t+k} - \hat{p}_{t+k} - \frac{1}{1 - \alpha}(\hat{a}_{t+k} - \alpha\hat{y}_{t+k}) - \frac{\alpha}{1 - \alpha}(\hat{y}_{t+k} - \hat{y}_{t+k|t}) \\ &= \hat{m}c_{t+k}^r - \frac{\alpha\varepsilon}{1 - \alpha}(\hat{p}_t^* - \hat{p}_{t+k}), \end{aligned} \quad (4.1.18)$$

where the last line uses 4.1.12.

Substituting 4.1.18 into 4.1.14, we have

$$\begin{aligned} p_t^* &= \Gamma(\hat{m}c_{t+k|t}^r + \hat{p}_{t+k}) \\ &= \Gamma\left[\hat{m}c_{t+k}^r - \frac{\alpha\varepsilon}{1 - \alpha}(\hat{p}_t^* - \hat{p}_{t+k}) + \hat{p}_{t+k}\right] \\ &= -\frac{\alpha\varepsilon}{1 - \alpha}\hat{p}_t^* + \Gamma(\hat{m}c_{t+k}^r + \Theta^{-1}\hat{p}_{t+k}), \end{aligned}$$

where $\Gamma(\cdot) \equiv (1 - \beta\theta) \sum_{k=0}^{\infty} (\beta\theta)^k \mathbb{E}_t(\cdot)$, and $\Theta = \frac{1 - \alpha}{1 - \alpha + \alpha\varepsilon} \leq 1$.

Solving for the optimal price in terms of average real marginal cost:

$$\begin{aligned} \hat{p}_t^* &= \Gamma(\Theta\hat{m}c_{t+k}^r + \hat{p}_{t+k}) \\ &= (1 - \beta\theta)(\Theta\hat{m}c_t^r + \hat{p}_t) + \beta\theta\mathbb{E}_t\hat{p}_{t+1}^*. \end{aligned} \quad (4.1.19)$$

Similarly, substituting 4.1.19 into 4.1.11, then subtracting by $(1 - \theta)\hat{p}_t$ and arranging yields

$$\hat{\pi}_t = \beta\mathbb{E}_t\hat{\pi}_{t+1} + \lambda\hat{m}c_t^r, \quad (4.1.20)$$

where $\lambda \equiv \frac{(1 - \theta)(1 - \beta\theta)}{\theta}\Theta$.

²We can prove that $d_t = \ln D_t$ is equal to zero up to a first-order approximation in a neighborhood of $\pi = 0$.

Solving forward, we obtain

$$\hat{\pi}_t = \lambda \sum_{k=0}^{\infty} \beta^k \mathbb{E}_t \hat{m}c_{t+k}^r.$$

Re-consider the log-linearized form of average real marginal cost, using 4.1.9 and 4.1.17 yields

$$\begin{aligned} \hat{m}c_t^r &= (\hat{w}_t - \hat{p}_t) - (\hat{a}_t - \alpha \hat{n}_t) \\ &= (\sigma \hat{y}_t + \varphi \hat{n}_t) - \frac{1}{1-\alpha} (\hat{a}_t - \alpha \hat{y}_t) \\ &= \sigma \hat{y}_t + \frac{\varphi}{1-\alpha} (\hat{y}_t - \hat{a}_t) - \frac{1}{1-\alpha} (\hat{a}_t - \alpha \hat{y}_t) \\ &= \left(\sigma + \frac{\varphi + \alpha}{1-\alpha} \right) \hat{y}_t - \frac{\varphi + 1}{1-\alpha} \hat{a}_t. \end{aligned} \quad (4.1.21)$$

Under flexible prices the real marginal cost is constant ($MC_t^r = \mu^{-1}$), thus $\hat{m}c_t^r = 0$. Defining the natural level of output, denoted by y_t^n , as the equilibrium level of output under flexible prices:

$$\hat{y}_t^n = \frac{\varphi + 1}{\varphi + \alpha + \sigma - \alpha\sigma} \hat{a}_t.$$

Then we have

$$\hat{m}c_t^r = \left(\sigma + \frac{\varphi + \alpha}{1-\alpha} \right) \tilde{y}_t, \quad (4.1.22)$$

where $\tilde{y}_t \equiv \hat{y}_t - \hat{y}_t^n$ is denoted as the output gap, the deviation between the real output and the natural level of output.

Substituting 4.1.22 into 4.1.20, we obtain the New Keynesian Philips Curve (*abbr.* NKPC):

$$\hat{\pi}_t = \beta \mathbb{E}_t \hat{\pi}_{t+1} + \kappa \tilde{y}_t, \quad (4.1.23)$$

where $\kappa \equiv \lambda \left(\sigma + \frac{\varphi + \alpha}{1-\alpha} \right) > 0$.

Rewriting 4.1.16 in terms of the output gap as

$$\tilde{y}_t = \mathbb{E}_t (\hat{y}_{t+1}^n - \hat{y}_t^n) + \mathbb{E}_t \tilde{y}_{t+1} - \sigma^{-1} (\hat{i}_t - \mathbb{E}_t \hat{\pi}_{t+1}).$$

Define the natural rate of interest $\hat{r}_t^n \equiv \sigma \mathbb{E}_t (\hat{y}_{t+1}^n - \hat{y}_t^n)$. Then rewriting the equation in terms of $\hat{r}_t^n$, we obtain the Dynamic IS Curve (*abbr.* DISC):

$$\tilde{y}_t = -\sigma^{-1} (\hat{i}_t - \mathbb{E}_t \hat{\pi}_{t+1} - \hat{r}_t^n) + \mathbb{E}_t \tilde{y}_{t+1}. \quad (4.1.24)$$

Under the assumption that the effects of nominal rigidities vanish asymptotically, $\lim_{T \rightarrow \infty} \mathbb{E}_t \tilde{y}_{t+T} = 0$. In that case one can solve 4.1.24 forward to yield

$$\tilde{y}_t = -\sigma^{-1} \sum_{k=0}^{\infty} \mathbb{E}_t (\hat{r}_{t+k} - \hat{r}_{t+k}^n),$$

where $\hat{r}_t^n \equiv \hat{i}_t - \mathbb{E}_t \hat{\pi}_{t+1}$ is the (log-linearized) expected real return on a one period bond (i.e., the real interest rate).

In the non-policy block of basic New Keynesian model, The NKPC determines inflation given a path for the output gap, and The DISC determines output gap given a path for the (exogenous) natural rate and the actual real rate. In order to close the model, we need to supplement these two equations with one or more equations determining how the nominal interest rate $\hat{i}_t$ evolves over time, i.e., with a description of how the monetary policy is conducted.

4.1.4 Monetary Policy

Consider the equilibrium under a simple interest rate rule of the form (as known as the Taylor rule)

$$\hat{i}_t = \phi_\pi \hat{\pi}_t + \phi_y \tilde{y}_t + v_t, \quad (4.1.25)$$

where v_t is an exogenous (possibly stochastic) component with zero mean. Assume ϕ_π and ϕ_y are non-negative coefficients, chosen by the monetary authority.

Combining 4.1.25 with 4.1.24, the DISC becomes

$$\tilde{y}_t = -\sigma^{-1}[\phi_\pi \hat{\pi}_t + \phi_y \tilde{y}_t - \mathbb{E}_t \hat{\pi}_{t+1} - (\hat{r}_t^n - v_t)] + \mathbb{E}_t \tilde{y}_{t+1}.$$

Rewriting the NKPC and DISC:

$$\begin{aligned} (1 + \sigma^{-1} \phi_y) \tilde{y}_t + \sigma^{-1} \phi_\pi \hat{\pi}_t &= \mathbb{E}_t \tilde{y}_{t+1} + \sigma^{-1} \mathbb{E}_t \hat{\pi}_{t+1} + \sigma^{-1} (\hat{r}_t^n - v_t), \\ -\kappa \tilde{y}_t + \hat{\pi}_t &= \beta \mathbb{E}_t \hat{\pi}_{t+1}. \end{aligned}$$

The matrix form becomes:

$$\begin{bmatrix} \tilde{y}_t \\ \hat{\pi}_t \end{bmatrix} = \mathbf{A}_T \begin{bmatrix} \mathbb{E}_t \tilde{y}_{t+1} \\ \mathbb{E}_t \hat{\pi}_{t+1} \end{bmatrix} + \mathbf{B}_T (\hat{r}_t^n - v_t), \quad (4.1.26)$$

where $\Omega \equiv (\sigma + \phi_y + \kappa \phi_\pi)^{-1}$, $\mathbf{A}_T \equiv \Omega \begin{bmatrix} \sigma & 1 - \beta \phi_\pi \\ \kappa \sigma & \kappa + \beta(\sigma + \phi_y) \end{bmatrix}$, and $\mathbf{B}_T \equiv \Omega \begin{bmatrix} 1 \\ \kappa \end{bmatrix}$.

By the Blanchard-Kahn condition, given that both the $\tilde{y}_t$ and $\hat{\pi}_t$ are nonpredetermined variables, the solution to 4.1.26 is locally unique, if and only if, $\mathbf{A}_T$ has both eigenvalues within the unit circle. Under the assumption of non-negative coefficients (ϕ_π, ϕ_y) it can be shown that a necessary and sufficient condition for uniqueness is given by

$$\kappa(\phi_\pi - 1) + (1 - \beta)\phi_y > 0, \quad (4.1.27)$$

which is assumed to hold, unless stated otherwise.

Then we consider the monetary policy shock in the system. Assume an AR(1) process of $\{v_t\}$, that is, $v_t = \rho_v v_{t-1} + \varepsilon_t^v$, $\rho_v \in [0, 1)$. A positive (negative) realization of ε_t^v should be interpreted as a contractionary (expansionary) monetary policy shock, leading to a rise (decline) in the nominal interest rate, given $\hat{\pi}_t$ and $\tilde{y}_t$. Because the natural rate of interest (output) is not affected by monetary shocks, $\forall t$, $\hat{r}_t^n = \hat{y}_t^n = 0$ is set for the purposes of the present exercise.

Using undetermined coefficient method, guess that

$$\begin{aligned}\tilde{y}_t &= \psi_{yv} v_t, \\ \hat{\pi}_t &= \psi_{\pi v} v_t,\end{aligned}$$

where ψ_{yv} and $\psi_{\pi v}$ are coefficients to be determined.

Verification:

$$\begin{aligned}\tilde{y}_t &= -(1 - \beta\rho_v)\Lambda_v v_t, \\ \hat{\pi}_t &= -\kappa\Lambda_v v_t,\end{aligned}$$

where $\Lambda_v \equiv [(1 - \beta\rho_v)[\sigma(1 - \rho_v) + \phi_y] + \kappa(\phi_\pi - \rho_v)]^{-1}$.

Since 4.1.27 is satisfied, it can be easily shown that $\Lambda_v > 0$. Then we have $-(1 - \beta\rho_v)\Lambda_v < 0$, and $-\kappa\Lambda_v < 0$. Hence, If $v_t \uparrow$, then $\hat{\pi}_t \downarrow$, $\tilde{y}_t \downarrow$, and $\hat{y}_t \downarrow$.

Similarly, we can solve the form of (log-linearized) real interest $\hat{r}_t \equiv \hat{i}_t - \mathbb{E}_t \hat{\pi}_{t+1}$ as

$$\hat{r}_t = \sigma(1 - \rho_v)(1 - \beta\rho_v)\Lambda_v v_t,$$

thus we have $\sigma(1 - \rho_v)(1 - \beta\rho_v)\Lambda_v > 0$.

Hence,

$$\hat{i}_t = [\sigma(1 - \rho_v)(1 - \beta\rho_v) - \kappa\rho_v]\Lambda_v v_t.$$

If $v_t \uparrow$, then $\hat{r}_t \uparrow$. But if the persistence of the monetary policy shock ρ_v is sufficiently high, $\hat{i}_t \downarrow$, since The downward adjustment in the nominal rate induced by the decline in inflation and the output gap more than offsetting the direct effect of a higher v_t .

Chapter 5

Incomplete Market Models

5.1 The Completeness of Market Structure

Definition 5.1.1: Arrow-Debreu Security

Let $\mathcal{S} = \{1, \dots, S\}$ be the finite set of possible future states. Denote the payoff of an Arrow-Debreu security for state s as d_s , then it is a function $d_s : \mathcal{S} \rightarrow \mathbb{R}$ defined by $d_s(s') = \mathbb{I}_{\{s'=s\}}$.

In a discrete state-space, we can represent the payoff of this security as a payoff vector $\mathbf{e}_s \in \mathbb{R}^S$. For the security corresponding to state s , the vector has a 1 in the s -th position and 0 elsewhere:

$$\mathbf{e}_s = (0, \dots, 0, 1, 0, \dots, 0)^T$$

Let there be J available assets in the market, the payoff of each asset j is represented by a vector $\mathbf{a}_j \in \mathbb{R}^S$. Define the payoff matrix A as:

$$A = (\mathbf{a}_1, \dots, \mathbf{a}_J)^T = \begin{pmatrix} a_{11} & \cdots & a_{1J} \\ \vdots & \ddots & \vdots \\ a_{S1} & \cdots & a_{SJ} \end{pmatrix} \in \mathbb{R}^{S \times J},$$

where a_{sj} represents the asset j in state s .

Definition 5.1.2: Complete Market

A market is complete if, $\text{span}(A) = \mathbb{R}^S$. That is equivalent to the rank condition of the payoff matrix $\text{rank}(A) = S$.

In a complete market, any arbitrary contingent claim can be perfectly replicated (synthesized) by a linear combination of existing assets (strictly, $\text{span}(A)$ is dense in L^2 space). There is no "unhedgeable" risk, and the equilibrium allocation is Pareto optimal. Moreover, we have proved that the optimal consumption allocation is not history-dependent when a exogenous Markov state variable determines the aggregate endowment.

But in a incomplete market, some Arrow-Debreu security do not works ($\text{rank}(A) = S$), so individuals may face idiosyncratic risk, and the history dependence is encoded in the dependence of a household's consumption on its current asset holdings. In addition, the incomplete markets model also can help to explain the “equity premium puzzle” that [Mehra and Prescott \(1985\)](#) showed.

5.2 Borrowing Constraint and Idiosyncratic Shocks

The incomplete markets model is first studied by [Bewley \(1980\)](#). Bewley's model establishes a foundational framework where agents face uninsurable idiosyncratic labor income shocks and borrowing constraints. Lacking state-contingent claims, individuals engage in self-insurance through asset accumulation.

Consider an agent's consumption-saving problem. In period t , let $c_t \geq 0$ denote the consumption, and $a_t \geq -b$ denote the asset holdings or borrowings with constraint b , and l_t denote the labor endowment shock. The agent takes the interest rate r and the wage rate w as given and solves the following problem:

$$\max_{c_t, a_{t+1}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t)$$

s.t.

$$\begin{aligned} c_t + a_{t+1} &= (1+r)a_t + wl_t, \\ a_t &\geq -b. \end{aligned}$$

Assume that l_t takes values on $[l_{\min}, l_{\max}]$ and follows a Markov process with transition function Q .

Moreover, we impose the transversality condition $\lim_{t \rightarrow \infty} \frac{a_t}{(1+r)^t} = 0$, which also implies the following natural borrowing constraint:

$$a_t \geq -\frac{wl_{\min}}{r},$$

which can be derived by non-negative constraints on consumption ($c_t \geq 0$).

In sum, the borrowing constraint can be represented as

$$\begin{aligned} a_t &\geq -\phi, \\ \phi &= \begin{cases} \min\{b, \frac{wl_{\min}}{r}\} & \text{if } r > 0, \\ b & \text{if } r = 0. \end{cases} \end{aligned}$$

Define the cash on hand or total resource x_t as

$$x_t \equiv (1+r)a_t + wl_t + \phi.$$

Let $\hat{a}_t = a_t + \phi$ and $y_t = wl_t$. Suppose y_t is IID, then the process x_t is defined by

$$x_{t+1} = R\hat{a}_{t+1} + y_{t+1} - r\phi = R(x_t - c_t) + y_{t+1} - r\phi.$$

The agent solves the following dynamic programming problem:

$$V(x) = \max_{c, x'} \{u(c) + \beta \mathbb{E}V(x')\},$$

s.t.

$$\begin{aligned} c + \hat{a}' &= x, \\ x' &= R(x - c) + y - r\phi. \end{aligned}$$

The FOC is given by

$$u'(x - \hat{a}') = \beta R \mathbb{E}V'(R\hat{a}' + y - r\phi),$$

where V' stands for the marginal utility of consumption.

Comparing to the case of complete market, now we prove that when $\beta R < 1$, there exists unique invariant distribution.

Lemma 5.2.1: Contraction Mapping

Let $\beta \in (0, 1)$ and $R > 0$. The consumption policy function $c(x)$ is strictly increasing for $x > x_{min}$ and the saving policy function $\hat{a}_x$ satisfies

$$0 \leq \hat{a}(x') - \hat{a}(x) < x' - x, \quad \forall x_{min} \leq x \leq x'.$$

Proof. Based on the FOC, we define a function $f(s; x)$ by

$$f(s; x) = u'(x - s) - \beta R \mathbb{E}V'(Rs + y - r\phi).$$

Thus, $s(x)$ is the transition function whenever $f(s(x); x) = 0$.

By the Euler equation at x and $x' > x$, since $u''(c) < 0$, we deduce that $f(\hat{a}(x); x') < f(\hat{a}(x); x) = 0$, which implies $\hat{a}(x') > \hat{a}(x)$ due to $f(\hat{a}(x'); x') = 0$, and f is increasing to s (or $\hat{a}(x)$). Hence, $0 \leq \hat{a}(x') - \hat{a}(x)$.

In addition,

$$\begin{aligned} f(\hat{a}(x) + x' - x; x') &= u'(x - \hat{a}(x) - (x' - x)) - \beta R \mathbb{E}V'(R\hat{a}(x) + R(x' - x) + y - r\phi) \\ &> u'(x - \hat{a}(x)) - \beta R \mathbb{E}V'(R\hat{a}(x) + R(x' - x) + y - r\phi) \\ &= \beta R \mathbb{E}V'(R\hat{a}(x) + y - r\phi) - \beta R \mathbb{E}V'(R\hat{a}(x) + R(x' - x) + y - r\phi) > 0. \end{aligned}$$

Combine with $f(\hat{a}(x); x') < f(\hat{a}(x); x) = 0$, by the intermediate value theorem we obtain

$$\hat{a}(x') \in (\hat{a}(x), \hat{a}(x) + x' - x),$$

that is $0 \leq \hat{a}(x') - \hat{a}(x) < x' - x$. □

Lemma 5.2.2: Binding Borrowing Constraint

Let $\beta R < 1$. Suppose that $x_{min} = y_{min} - r\phi > 0$. Then $\exists \hat{x} > x_{min}$ s.t. if $x_t \leq \hat{x}$, then $\hat{a}_{t+1} = 0$, and if $x_t > \hat{x}$, then $\hat{a}_{t+1} > 0$.

Lemma 5.2.3: Maximum Wealth Level

Let $\beta R < 1$. If $-\frac{cu''(c)}{u'(c)} < \gamma$ for all c sufficiently large for some $\gamma > 0$, then $\exists x^*$, s.t. if $x_t > x^*$, then $x_{t+1} < x_t$.

5.3 Incomplete Market in Pure Exchange Economy**5.4 Incomplete Market in Production Economy****5.5 Aggregate Shocks and Approximate Aggregation**